Anonymization and Information Loss

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Abstract

Anonymizing financial texts is a widely used technique to reduce look-ahead bias when extracting textual information using large language models. We demonstrate that while anonymization effectively obscures firm identity, it compromises textual understanding, diminishing models' ability to extract meaningful economic signals from financial texts. This information loss is particularly severe when numerical and agent entities are removed, and is amplified in texts with high linguistic uncertainty, high entity content, or low firm transparency. We find this adverse effect to be pervasive across various tasks, including the prediction of firm uncertainty, corporate policy, and default. Furthermore, we show that this information loss is widespread across different large language models and across various types of financial texts. Notably, in sentiment extraction from earnings call transcripts, the information loss induced by anonymization proves more severe than the effects of look-ahead bias, suggesting that the costs of anonymization may outweigh its benefits in certain financial applications.

Keywords: Large Language Models, Look-ahead Bias, Memorization, Anonymization, Information Loss, Textual Analysis

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1 Introduction

Large language models (LLMs) have transformed how finance and accounting researchers extract information from textual data. Numerous studies underscore the power of LLMs compared to traditional methodologies like bag-of-words (e.g., Lopez-Lira and Tang, 2023). Researchers use LLMs to extract economic signals from earnings calls, interpret firm policies, and measure geopolitical exposure (Jha et al., 2024a,b; Siano, 2025; Kim et al., 2023; Clayton et al., 2025). Other applications include mapping global production networks using financial texts (Breitung and Müller, 2025), detecting investor sentiment on social media (Chen et al., 2025), and generating structured economic representations from news (Sarkar, 2025).

Despite their power in analyzing financial texts, several studies caution that LLMs may introduce look-ahead bias in the data extraction process. This bias is particularly relevant for finance and economics, where both extracted signals are leveraged to predict future economic performance or stock returns. This bias stems from LLMs' training stage, when they learn from vast corpora spanning long periods of time, potentially causing their outputs to incorporate information that would not have been accessible to an agent prior to the model's knowledge cutoff (e.g., Levy, 2024; Sarkar and Vafa, 2024; Ludwig et al., 2025; Lopez-Lira et al., 2025).

Several strategies have been proposed to mitigate this bias. Among them, anonymization has gained significant popularity since it is relatively easy to implement while still allowing researchers to use state-of-the-art models. This methodology replaces identifiers such as firm names or tickers with placeholders (e.g., "FIRM_1") (Glasserman and Lin, 2023; Engelberg et al., 2025). The premise is that obscuring these unique identifiers forces LLMs to analyze the document exclusively on its inherent linguistic and semantic content rather than firm-specific knowledge.[1]

[1] Other approaches include training chronologically consistent models that only use data before the date of the textual content of interest (e.g., Sarkar, 2024; He et al., 2025) or conducting robustness tests using texts beyond existing models' knowledge cutoff (e.g., Jha et al., 2024a; Siano, 2025).

In this study, we investigate an important limitation of the anonymization methodology: its potential to lead to significant information loss and degrade the extracted signal. The identity of an entity is not merely a label but a critical piece of contextual information essential for accurate interpretation. Consider the statement: "The firm announced it will lay off 10,000 employees." For a company undergoing financial distress, such as a struggling retailer facing bankruptcy, this announcement likely reinforces a negative outlook, signaling deepening trouble. Conversely, for a highly profitable technology giant undertaking strategic restructuring, the same announcement could be interpreted positively by the market as a sign of disciplined cost management. After anonymization, both scenarios collapse into an identical, ambiguous input: "FIRM_1 announced it will lay off NUMBER_1 employees." This procedure strips the model of the necessary context to make an accurate inference, potentially leading to noisy signals.

Empirically, we task LLMs to extract information from a sample of conference call transcripts that are dated after the models' knowledge cutoff. We compare the informativeness of signals extracted before and after popular anonymization procedures. As these conference call transcripts are not part of the models' pretraining datasets, any observed differences in the extracted signals' informativeness can be attributed directly to the changes in information content within the transcripts introduced by anonymization.

Our first analysis focuses on the extraction of firm-value-relevant signals using GPT-4o-mini. We prompt the model to classify whether a given conference call contains positive or negative sentiment regarding future firm value. Subsequently, we analyze the relationship between the extracted signal and short-term abnormal returns around the conference call date. This setting closely aligns with past research designs (e.g., Loughran and Mcdonald, 2011; Garcia et al., 2023; Siano, 2025).[2] An effective signal is expected to exhibit a stronger relationship with short-term abnormal returns.

[2]Our analyses are different from Lopez-Lira and Tang (2023); Chen et al. (2022); Engelberg et al. (2025), who focus on predicting future returns using LLM-extracted signals. Predicting future returns may require investors to underreact to value-relevant information. This requirement could confound the goal of evaluating LLMs' ability to extract textual sentiment.

We find that sentiment extracted from raw (i.e., unanonymized) earnings call transcripts exhibits a strong association with conference call-day stock returns, indicating the model's capacity to capture value-relevant information. Subsequently, we anonymize transcripts using two strategies. The first algorithm is from the spaCy library, a powerful toolkit for identity recognition, and is designated as Transformer (TRF), which is the most capable NER model in the spaCy library. We also consider the anonymization approach analyzed in Engelberg et al. (2025), which uses LLMs to iteratively mask entities and paraphrase the text. We find that the signals extracted from anonymized texts exhibit significantly attenuated informativeness. For instance, when using a standard Transformer-based anonymization model, the R^2 from a regression of returns on sentiment with controls drops by over 6% (from 0.132 to 0.124) and even drops by more than 10% (from 0.078 to 0.070) in the regression without controls. The information loss becomes even clearer in a direct "horse race" regression. When both raw and anonymized sentiment signals are included as explanatory variables, the coefficient on the raw signal remains large and highly significant, while the coefficient on the anonymized signal collapses from 2.331 to 0.775. This suggests that the sentiment information conveyed by the anonymized signal is largely subsumed by the information provided by the named entities.

We trace the source of this information loss to the removal of specific entity types. We classify entities into four main types: NUMBERS, AGENTS, PLACES, and OTHERS. The degradation in signal quality is most severe when removing numbers (such as financial figures and dates) and agents (such as company and executive names), which act as critical anchors for contextual interpretation. In contrast, removing entities like places has a minimal impact.

Furthermore, we find that the divergence between raw and anonymized signals is greater in texts characterized by higher linguistic uncertainty, or those concerning smaller, more specific firms with less public exposure. This suggests that in ambiguous contexts, LLMs rely more heavily on concrete named entities to ground their analysis, making their removal particularly impactful. For these firms, the model lacks broad contextual knowledge, forcing it to rely disproportionately on specific named entities to infer meaning. Consequently, removing these entities causes more severe information loss

While our main analyses focus on GPT-4o-mini, we show that bigger models (e.g., GPT-4o) and reasoning-focused models (e.g., GPT-o3-mini) also exhibit information loss after conference calls are anonymized, indicating the generalizability of our results. In fact, we find that the information loss from anonymization is more pronounced when conducting information extraction using bigger models.

The loss of information associated with anonymization is also not confined to extracting signals related to firm value. We attempt to extract signals relating to uncertainty and find that anonymization leads to a significant reduction in the ability to predict future return volatility. Furthermore, we follow Jha et al. (2024a) and Jha et al. (2024b) to predict future corporate policy and firm performance. We find that anonymization similarly degrades models' ability in those prediction tasks.

Moreover, to confirm the generalizability of our conclusions, we replicate the sentiment analysis tests on a different corpus with distinct characteristics. Specifically, we task LLMs to decipher the sentiment signals embedded in short, entity-dense news headlines. The results are consistent, confirming that the loss of information is a fundamental consequence of the anonymization process itself, rather than a specific type of corporate disclosure.

Finally, we extend our analysis to include the pre-cutoff period (November 2022–October 2023), allowing us to compare information loss across periods where look-ahead bias might differentially affect LLM performance. In the context of explaining contemporaneous stock returns, we find limited evidence of look-ahead bias, as the performance gap between raw and anonymized signals remains stable across both pre- and post-cutoff periods.

Our study contributes to the emerging literature on LLM-based information extraction by addressing a critical methodological trade-off. While current research cautions against look-ahead bias (Glasserman and Lin, 2023; Sarkar and Vafa, 2024) and proposes anonymization as a remedy (Breitung and Müller, 2025; Engelberg et al., 2025), our findings highlight a significant hidden cost. We demonstrate that while anonymization may mitigate bias, it does so at the expense of essential contextual value. Consequently, our results urge caution: without a post-cutoff sample for comparison, researchers may mistakenly attribute a reduction in signal informativeness to the successful mitigation of bias, when it is actually a byproduct of information loss.

2 Methodology and Data

2.1 Anonymization Algorithms

We employ two approaches for text anonymization. The first utilizes a specialized Named Entity Recognition (NER) model from the Python spaCy package: en-core-web-trf (TRF), which is based on a Transformer architecture. The second approach follows Engelberg et al. (2025) and uses a commercial LLM, specifically GPT-4o-mini, in conjunction with custom prompts to perform anonymization.

These two models vary significantly in size, a difference that stems from their entirely different underlying architectures. The en-core-web-trf model is based on the RoBERTa-base Transformer architecture, containing approximately 0.1 billion parameters.[3] In contrast, the GPT-4o-mini is larger by several orders of magnitude; while its exact parameters are undisclosed, it is widely estimated by the industry to be on the order of 8 billion parameters.[4]

Approach A in Figure 1 illustrates the workflow for the spaCy NER model (*en-core-web-trf*). The model first scans the raw text to recognize specific entities, such as "Apple," "Tim Cook," and "Luca Maestri" in the example shown. It then generates a consistent global map based on the entity type and order of appearance—for instance, mapping "Apple" to ORG_1 as the first organization identified, and "Tim Cook" to PERSON_1 as the first person.[5]

[3] https://huggingface.co/roberta-base

[4] https://techcrunch.com/2024/07/18/openai-unveils-gpt-4o-mini-a-small-ai-model-powering-chatgpt

[5] As exhibited in Table A.4, we group these 18 types into four main categories to structure our anonymization: NUMBERS (which includes DATE, CARDINAL, MONEY, PERCENT, ORDINAL, TIME, and QUANTITY), PLACES (GPE, LOC, FAC), AGENTS (PERSON, ORG, NORP), and OTHERS (PROD-

Finally, the system applies these mappings globally to the entire text unit, producing an anonymized output that preserves textual structure while obscuring identities.

Approach B in Figure 1 describes the strategy using Large Language Models (specifically GPT-4o-mini). Unlike the linear NER approach, this method employs an iterative verification loop. The model first attempts to mask entities by replacing them with placeholders. The system then evaluates whether the entities in this masked output can still be identified by an LLM. If the guess matches the actual name (based on a string similarity threshold), the system loops back to the masking stage. Once the text passes this initial check, it undergoes a paraphrasing process where sentence structures are rewritten to remove stylistic identifiers (e.g., changing "Speaking first today..." to "We will begin today with remarks from..."). A final identification check is performed on this paraphrased output.[6] If the firm or agents can still be identified, the process restarts from the beginning. This cycle of masking and paraphrasing repeats for a maximum of three rounds, as the marginal benefit of anonymization diminishes significantly beyond this point. Only when the text passes the final verification or reaches the round limit is the final LLM-based anonymization text generated. Our prompt design follows Engelberg et al. (2025), and the full prompt is available in Appendix B.1.

Table 1 presents a text excerpt from Apple Inc.'s 2024 Q1 earnings call transcript, along with the anonymization results from the TRF and LLM approaches. The TRF model exhibits a rigid, category-based replacement strategy; for instance, it anonymizes temporal expressions like "today" and "annual" into "DATE_1" and "DATE_2," and correctly identifies "SEC" as an organization ("ORG_2"). In contrast, the LLM approach offers a distinct progression. After several rounds of masking and paraphrasing, it demonstrates superior flexibility by altering the sentence structure itself like changing "Please note" to "Please be aware", and replacing specific identifiers with semantic descriptions, such as converting "SEC" into "the regulatory body," thereby mitigating identification risks associated with

UCT, EVENT, LAW, WORK_OF_ART, and LANGUAGE).

[6]We consider the text to be "identified" if the LLM's guessed firm name has a `fuzz.token_sort_ratio` score of 75 or higher when matched against the actual firm name, or if the estimated conference call date falls within 7 days of the actual date.

specific stylistic patterns.

2.2 Data

2.2.1 Stock Data

The sample for this research consists of constituent stocks from the iShares Russell 3000 ETF for the period of November 2023 through December 2024. We obtain return data from CRSP, corporate accounting data from Compustat, and EPS forecast data from the I/B/E/S Detail History database. Detailed variable definitions can be found in Table A.2.

Our sample of earnings call transcripts comprises 7,382 transcripts from 1,831 unique firms over a period of 259 days, as detailed in Table A.3. The sample firms are tilted toward large-cap firms, with an average market capitalization of approximately $23,845 million, an average book-to-market ratio of 0.55, and an average turnover rate of 2.77. 47.41% of these firms are listed on NASDAQ.

2.2.2 Earnings Call Transcripts

The primary textual source we examine in this study is corporate conference calls. They have been shown to carry rich value-relevant information about firms (e.g., Frankel et al., 1999; Brown et al., 2004). Many recent studies leverage LLMs to extract information from these texts (e.g., Jha et al., 2024a; Clayton et al., 2025). To isolate a clean effect unconfounded by look-ahead bias, we utilize data generated after the knowledge cutoff date of the Large Language Models (LLMs) employed. Models used in this study include GPT-4o-mini-2024-07-18, GPT-4o-2024-08-06, and GPT-o3-mini-2025-01-31, all of which have a knowledge cutoff date of October 2023[7]. Accordingly, we download earnings call transcripts from November 2023 onward from SeekingAlpha.

We remove duplicate entries and those with wrong time stamps. We also exclude transcripts exceeding 15,000 tokens. This threshold was established because the GPT-4o-mini

[7] https://github.com/HaoooWang/llm-knowledge-cutoff-dates

model, used for the anonymization procedure, has a maximum output limit of 16,000 tokens; longer input transcripts would therefore result in truncated anonymized text. Furthermore, we match the transcript data with corresponding returns and EPS forecast errors using date and PERMNO. The detailed sample selection criteria are presented in Panel A of Table A.1.

To facilitate our analysis of the pre-cutoff period, we extend our dataset to include earnings call transcripts from November 2022 to October 2023. We apply the same filtering criteria to this earlier sample, and match observations with stock returns and EPS forecast errors. To ensure a consistent comparison across the pre- and post-cutoff periods and to mitigate potential selection bias, our combined analysis is restricted to a balanced panel of firms that have valid observations in both the pre-cutoff (November 2022–October 2023) and post-cutoff (November 2023–December 2024) periods.

2.2.3 News Headlines

Besides conference calls, news articles have also been shown to deliver rich information about firms (e.g., Tetlock et al., 2008; Tetlock, 2010). Moreover, many studies also attempt to extract signals from news headlines and articles (e.g., Bybee et al., 2023; Lopez-Lira and Tang, 2023; Engelberg et al., 2025). To ensure the generalizability of our results, we also collect daily news headlines from the Finnhub API. Consistent with the out-of-knowledge-cutoff principle, our data collection begins on November 1, 2023. Following the methodology of Ke et al. (2019), we first exclude headlines associated with multiple firm tickers. To handle duplicate articles that arise from re-syndication across different news outlets, we retain news only from the most frequent source in our dataset, which is Yahoo Finance. Furthermore, we remove headlines containing the word "stock" or "stocks," as these titles may directly allude to stock price movements or trading volume, thus contaminating the signal.

We retain only headlines that feature a single organization. This is necessary because when we prompt the GPT model to assess the price implications for "the corresponding firm," headlines mentioning multiple or no specific firms would create ambiguity and confuse the model. Additionally, guided by Bybee et al. (2023), we exclude articles published on weekends. Finally, the filtered headlines are matched with other relevant stock and corporate financial data. The complete sample filtering procedure is detailed in Panel B of Table A.1.

2.2.4 LLM-Generated Measurements

We design a suite of prompts to extract several measures from the aforementioned textual data, including sentiment, uncertainty, investment (the firm's outlook on future capital expenditures), economy (the firm's outlook on the future macroeconomy), and the probability of credit rating downgrade. For each measure, we generate a quantitative score. The specific definitions for these scores and the complete prompts are available in the Appendix B.3 to B.8.

For the earnings call transcripts, we aggregate these scores by PERMNO and date. The matching rule is as follows: for calls held before the U.S. market close (16:00 ET), the announcement date (date 0) is set to the same calendar day. For calls held after the market close, the announcement date is set to the next trading day. This convention ensures that the stock return for the announcement date fully incorporates the information disclosed in the transcript.

Since a single stock may have multiple headlines on any given day, we average the scores of all headlines for each PERMNO-date to create a daily score. We apply the same rule. Specifically, daily average scores are categorized as pre- or post-market-close based on publication times and are assigned an announcement date using the identical rule as for conference call transcripts.

3 Results

3.1 Anonymization Performance in Earnings Calls

We examine the effectiveness of various anonymization algorithms by analyzing whether LLMs can recognize firm and date entities from the anonymized texts. First, we analyze the number of entities that were identified by anonymization algorithms. To achieve this goal, we tokenize these call transcript data using the Xenova/GPT-4o tokenizer.[8] We then use two anonymization algorithms to identify entities in conference calls. In Panel A of Table 2, we report the conference call transcript characteristics. The average transcript length is 8,895 tokens, with 25th and 75th percentiles at 6,971 and 10,901 tokens, respectively. We also report the percentage of identified entities based on each anonymization algorithm. Because the LLM generates plain text rather than structurally tagged tokens, we calculate the entity percentage by measuring the proportion of anonymized placeholders in the final output for both methods. This ensures the metrics are comparable.

Next, we examine the effectiveness of various anonymization algorithms. Following past literature (e.g., Engelberg et al., 2025), we task large language models (LLMs) with identifying masked date, firm name, Ticker and industry identities of anonymized transcripts. Specifically, GPT-4o-mini attempts to deanonymize the data using the prompt detailed in the Appendix B.2, and we report these entity identification results in Panel B.

We find that the TRF algorithm exhibits weaker anonymization performance. GPT-4o-mini successfully identifies 62.71% of firm names and 50.26% of tickers, although date identification remains low at 5.39%. In contrast, anonymization through the LLM-based approach proves to be significantly more thorough. Consistent with Engelberg et al. (2025), the LLM-based method drastically reduces re-identification rates, leaving only 5.49% of firm names and 3.09% of dates identifiable. This underscores the superior capability of the

[8] https://huggingface.co/Xenova/gpt-4o

iterative LLM approach in obscuring firm-specific information compared to the standard NER-based substitution.

3.2 Sentiment Extraction

Our primary analysis investigates the impact of anonymization on the informational content of financial text. We extract the sentiment from transcripts using GPT-4o-mini for the original transcript, as well as the anonymized ones. We design a prompt that closely follows Engelberg et al. (2025). The prompt used to extract sentiment is detailed in the Appendix B.3. We tabulate the pairwise correlation between the extracted sentiment scores in Table 3. We find that the correlation between the sentiment score extracted from the raw and the TRF-anonymized transcripts is 0.864, while the correlation with the LLM-anonymized transcripts is 0.858, indicating that these signals remain highly correlated. Since the correlation between the anonymized and the raw score is not perfect, it potentially suggests the loss of information from the anonymization process.

To formally test the severity of the information loss, we examine the association between sentiment scores, extracted from earnings call transcripts, and contemporaneous DGTW-adjusted stock returns ($DGTW_{i,t}$, as defined by Daniel et al. (1997)). Using the contemporaneous returns as the dependent variable is consistent with prior literature (e.g., Loughran and Mcdonald, 2011; Siano, 2025). A reduced relationship after anonymization would demonstrate the information loss resulting from anonymization.[9]

Empirically, we adopt the following regression model:

$$DGTW_{i,t} = \beta \text{Sentiment}_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,t} \quad (1)$$

The dependent variable is the announcement-day DGTW-adjusted stock return. The key independent variables are the sentiment scores extracted from the raw text, as well as

[9]The use of contemporaneous stock returns is motivated by the fact that, compared to future returns, they capture the most complete and immediate market reaction to the sentiment expressed in the transcripts.

from the texts anonymized using TRF and LLM methods, respectively. Following (Tetlock et al., 2008; Engelberg et al., 2025), we include a standard set of control variables, $X_{i,t}$, in our model. These comprise analyst forecast error (FE), which prior studies have consistently shown to have significant explanatory power for abnormal returns (e.g., Bernard and Thomas, 1989; Hirshleifer et al., 2009), lagged DGTW-adjusted returns ($DGTW_{i,t-1}$, $DGTW_{i,t-2}$, $DGTW_{i,t-3}$, $DGTW_{i,t-22,t-4}$, and $DGTW_{i,t-253,t-23}$), and firm characteristics including the logarithm of firm size ($\ln(Size)$), the book-to-market ratio ($\ln(BM)$), and turnover ($\ln(Turnover)$). All regressions include date fixed effects. All independent variables are standardized, facilitating a direct comparison of coefficients across different text treatments and models.

In our regression analysis with controls (reported in Table 4), the sentiment score from the raw text, $Sentiment_{RAW}$, exhibits a strong and highly significant relationship with the contemporaneous stock return. As reported in column 1, the coefficient on the raw signal is 2.487 ($t = 18.832$), confirming that LLMs effectively capture sentiment signals. As shown in columns 2 and 3, sentiment scores from anonymized texts (TRF and LLM) remain significantly associated with returns. However, their coefficients are attenuated to 2.331 and 2.329, respectively. This is accompanied by a decline in adjusted R^2 values from 0.132 to 0.124, indicating information loss across both anonymization methods.

To more formally compare the power of sentiment scores, we conduct pairwise regressions that include both raw and anonymized sentiment scores (see Columns 4 and 5). We find the coefficient of $Sentiment_{RAW}$ remains large and highly significant, whereas the coefficients on the anonymized sentiment scores contract sharply. For example, the $Sentiment_{TRF}$ coefficient plummets from 2.331 to 0.775 (see Column 4), representing a 67% informational loss. Similarly, the $Sentiment_{LLM}$ coefficient drops to 0.798 (Column 5). These results indicate that the sentiment-relevant information in anonymized text is largely subsumed by that of the original text, providing strong evidence that entity removal causes a significant loss of sentiment-relevant information. As reported in Table A.6, our results are also robust without including control variables.

3.3 The Role of Different Types of Entities

While anonymization invariably results in information loss, the extent of this loss is likely heterogeneous across different types of information. As shown in Appendix Table A.4, entities such as NUMBERS and AGENTS constitute a much larger proportion of our financial text corpus than PLACES and OTHERS. The varying roles and frequencies of different named entity types in the text suggest that their informational value may differ. Thus, we hypothesize that the degree of information loss from anonymization depends on the category of the entity being replaced. We anticipate that replacing core entities directly linked to firm fundamentals and quantitative performance, such as NUMBERS and AGENTS, will result in a significantly greater loss of information than replacing entities that primarily provide contextual information, such as PLACES and OTHERS. To systematically test this hypothesis and identify the primary sources of information loss, we conduct the following analysis.

First, we mask each entity type, and task GPT-4o-mini to generate sentiment scores. In Table A.5, we tabulate the correlation of the sentiment scores extracted from the raw, fully anonymized, and partially anonymized transcripts. We find that the correlation between the raw and fully anonymized sentiment scores is lowest, while the correlation between the raw and place-masked sentiment scores is highest.

We further conduct horse race regression tests in Table 5 to identify the source of this information loss by selectively replacing different entity categories. Columns 1-5 present the regression results for the original text and for texts where NUMBERS, PLACES, AGENTS, and OTHERS entities have been replaced. We observe a decline in both the regression coefficients and the adjusted R^2 values when the first three categories are replaced, indicating a tangible loss of information. Conversely, replacing the OTHERS category results in a slight increase in the coefficient and R^2.[10] The results from the horse race regressions (reported in Columns 6-9) reveal that the sources of information loss are not uniformly distributed. When only PLACES entities are replaced (see Column 7), the sentiment score coefficients for the original versus the anonymized text are the closest (1.397 vs. 1.158), suggesting that the replacement of location information has the minimal impact on overall explanatory capability. In contrast, the information loss is most severe when replacing NUMBERS entities (reported in Column 6) and AGENTS entities (reported in Column 8). In these cases, the coefficient for the original text's sentiment score is substantially larger than that of the anonymized text. This unveils an important mechanism: in a financial context, numbers that directly quantify performance, scale, and expectations, as well as names directly associated with companies and key individuals, serve as the core information carriers for an LLM's assessment of market sentiment and corporate outlook.

3.4 The Role of Textual and Firm Characteristics

In our previous analyses, we established that anonymization leads to a significant loss of explanatory and predictive value, primarily attributable to the removal of numerical and agent-class entities. We next investigate the textual and firm characteristics that amplify or mitigate this information loss.

We hypothesize that the information loss caused by anonymization varies with both the linguistic properties of the text and the information environment of the firm. Specifically, we posit that the information loss will be magnified in texts characterized by high linguistic uncertainty and high entity density, as these contexts rely heavily on specific details for interpretation. Conversely, we hypothesize that the loss will be mitigated for firms with high information transparency (large size and high analyst coverage), as LLMs likely possess stronger prior knowledge of these firms from pre-training corpora. Finally, we expect firms

[10] A potential explanation for this is that the LLM's information extraction from the original text may be overly sensitive to irrelevant contextual details. By removing a small amount of information, this distraction is mitigated, leading to a marginal improvement in explanatory power.

with high product specificity to experience greater information loss, as we expect sparse information from LLMs' training data to prevent the models from fully understanding the firms' background after entity removal.

To test these hypotheses, we measure the loss of information for each press release by the absolute difference between the two sentiment scores ($|Sentiment_{RAW} - Sentiment_{TRF}|$). This measure directly captures the magnitude of the "disagreement" in the LLM's judgment caused by the anonymization process. Table 6 presents the results of a series of univariate regressions using this divergence as the outcome.

The first textual predictor is the uncertainty score derived from the original transcript ($Uncertainty_{RAW}$); this metric is constructed using a prompt based on the definition by Loughran and Mcdonald (2011), which prioritizes linguistic imprecision over narrow risk-based measures (see Appendix B.4). To characterize the firm's information environment, we incorporate traditional measurements including the logarithm of market capitalization ($ln(Size)$) and analyst coverage ($Coverage$). We also consider product market competition using the TNIC similarity index ($TNIC3TSIMM$). Furthermore, we include the proportion of entities successfully identified by the TRF model (*Entity*%), which serves as a direct proxy for the volume of information removed. To ensure the comparability of coefficients, all independent variables are standardized to a zero mean and unit variance.

The empirical results indicate that textual uncertainty is a primary driver of the divergence in sentiment judgments. As reported in Column 1 of Table 6, the uncertainty score derived from the raw text exhibits a positive and statistically significant correlation with the sentiment gap ($\beta = 0.040, t = 12.080$). Notably, the adjusted R^2 for this specification (0.042) is substantially higher than those of the other explanatory variables, suggesting that uncertainty carries the highest explanatory power for the observed sentiment shifts.

These findings suggest that when management discourse is characterized by low linguistic ambiguity, the sentiment signal remains robust to anonymization; in such cases, the removal of specific entities does not materially alter the LLM's evaluative framework. Conversely, in highly ambiguous, cautious, or conditional contexts, the specific identity of an entity provides critical "anchors" that assist the model in resolving sentiment. The anonymization process severs these contextual links, forcing the LLM to rely on a semantically sparser narrative. This results in a greater deviation from the original sentiment signal, implying that the informational value of entities is magnified under conditions of high uncertainty.

In addition to textual characteristics, we find that a firm's information environment significantly moderates the effect of anonymization. Since models are trained using large corpora from the internet, they inherently possess significant knowledge about informationally transparent firms. Consequently, we find that anonymization has a relatively mild impact on these firms. Column 2 indicates that firm size, measured as $\ln(\mathit{Size})$, is negatively and significantly associated with the sentiment score difference. This suggests that the extensive representation of larger firms within the LLM's pre-training corpora provides the model with a developed prior understanding, making its sentiment assessments more robust to identifying information loss.

Similarly, Column 3 reports a negative and significant coefficient for analyst coverage, corroborating the transparency hypothesis. Higher analyst coverage corresponds to increased public discourse, reducing the sparsity of the firm's representation in the LLM's training data. To quantify this, we follow He and Tian (2013), calculating the average number of monthly earnings forecasts over the 12 months preceding the 2021 fiscal year-end. The results confirm that greater informational transparency leads to a smaller sentiment gap.

We further examine the role of firm heterogeneity using the total product similarity metric. As outlined in our hypothesis, firms with unique product offerings present a dual challenge: they are discussed less frequently in public discourse, and their distinct market positioning renders their operational narratives more idiosyncratic. We proxy for this using the $TNIC3TSIMM$ index (Hoberg and Phillips, 2016), where lower values signify a more unique product portfolio. Our empirical evidence in Column 4 supports this hypothesis; the coefficient for $TNIC3TSIMM$ is negative and statistically significant, confirming that firms with unique product market characteristics experience a greater deviation in sentiment scores following anonymization.

Finally, the proportion of a transcript composed of named entities is a significant factor. As shown in Column 5, a higher percentage of entities is positively associated with the sentiment gap. This aligns with the intuition that a greater volume of entities corresponds to a larger amount of information being removed during the anonymization process, naturally resulting in a larger sentiment deviation.

3.5 Alternative LLMs

To test whether our findings are contingent on a specific LLM, we further conduct the sentiment extraction task using two alternative LLMs: GPT-4o and GPT-o3-mini. Compared to the workhorse model in our prior analyses (i.e., GPT-4o-mini), GPT-4o is a larger model with more parameters. It also exhibits a much stronger ability in textual tasks.[11] GPT-o3-mini is a new class of "reasoning model" that is designed to address complicated reasoning problems. While it is unclear if financial textual analysis involves complex reasoning, we include this model to ensure that our prior results remain relevant for this new class of models.

We repeat the analyses conducted in Table 4 using the two alternative models. We report our results in Table 7. We also repeat our baseline results using GPT-4o-mini as a reference (Columns 1, 4, and 7 repeat the Columns 1, 2, and 4 in Table 4). First, across all three models, the sentiment score derived from the original text consistently demonstrates significant explanatory power for stock returns (Columns 1-3). Second, when the analysis is performed on the TRF-anonymized text (Columns 4-6), we observe a uniform and marked decline in both the regression coefficients and the adjusted R^2. This indicates that information loss from anonymization is a widespread and robust phenomenon, detectable by all tested LLMs.

[11]Based on the assessments provided by OpenAI, GPT-4o scores 88.7 on the MMLU benchmark, while GPT-4o-mini obtains 82.0. See https://openai.com/index/GPT-4o-mini-advancing-cost-efficient-intelligence/

Notably, the magnitude of this impact varies across models, indicating that different LLMs exhibit different sensitivities to information loss when processing the same anonymized text. The GPT-4o model is the most severely affected; in the respective regression, it loses approximately 20% of the sentiment's explanatory power, with its coefficient dropping from 2.528 to 2.005 and its adjusted R^2 decreasing from 0.131 to 0.108. Despite these variations in magnitude, the directional finding of information loss remains unequivocal. Importantly, the horse race regressions (Columns 7-9) provide the most decisive evidence. In this direct comparison, the sentiment coefficient for the raw text consistently and significantly outweighs that of the anonymized text, irrespective of the evaluation model. We therefore conclude that the observed information loss is a fundamental consequence of the anonymization process itself, rather than an artifact of a specific model's analytical limitations. In conclusion, our findings remain robust to the choice of the underlying LLM.

3.6 Can Anonymization Affect the Extraction of Other Information?

To comprehensively assess the breadth and economic significance of information loss induced by anonymization, our analysis must extend beyond the explanation of contemporaneous stock returns. If named entities are indeed primary carriers of information within the text, their removal should also impair the text's ability to predict a firm's future fundamentals and risk. This section broadens our investigation to four key domains to verify the pervasiveness of this information loss: future stock volatility, future capital investment, future corporate operating performance, and future credit rating downgrades. Table 8 shows all regression results.

First, prior research establishes a positive link between the textual uncertainty in 10-K reports and subsequent stock return volatility (Loughran and Mcdonald, 2011). Inspired by their findings, we investigate whether anonymization impairs the earnings call transcript text's ability to predict future stock volatility. Column 1 of Table 8 examines the predictive power of an Uncertainty score extracted from the text for future realized volatility over the subsequent 22 trading days (Vol_{post}). The uncertainty score variable is the same as that used in Section 3.4. It is extracted using a prompt designed based on the definition from Loughran and Mcdonald (2011). In our empirical specification, we follow Loughran and Mcdonald (2014) and choose pre-announcement volatility, alpha, absolute abnormal returns, the natural logarithm of the firm's market capitalization, and the book-to-market ratio as control variables. A detailed definition is provided in Table A.2.

As shown in Column 1, we include both $Uncertainty_{RAW}$ and $Uncertainty_{TRF}$ scores in a horse-race regression, the coefficient on $Uncertainty_{RAW}$ (0.026) remains significant, whereas the coefficient on $Uncertainty_{TRF}$ (0.012) becomes statistically insignificant ($t = 0.777$). The result is consistent with the findings of (Loughran and Mcdonald, 2011) that more uncertain language from management is associated with higher future stock price volatility. And more importantly, this result suggests that once specific entities such as project names, figures, and personal names are redacted, the LLM's ability to infer future uncertainty is severely weakened, rendering its signal value negligible in the presence of the signal from the original text.

Next, we turn our focus to corporate physical investment decisions, following the approach of Jha et al. (2024a). We extract corporate expectations regarding future investment (Investment) from earnings call transcripts. Column 2 compares the predictive power for future investments of Investment scores from RAW and TRF-anonymized management's forward-looking statements on capital expenditures, with the dependent variable being capital expenditures in quarter $t+2$ (two quarters ahead). The results show that the coefficient on $Investment_{RAW}$ (0.043) is larger than that on $Investment_{TRF}$ (0.033), with both retaining high statistical significance. The observed decline in the coefficient's magnitude is consistent with the findings of Jha et al. (2024a). This indicates that the anonymized text still contains valid information about future investment trajectories, but that information loss is also present.

Subsequently, we examine the predictive power of textual information for a firm's core future operating performance, drawing on Jha et al. (2024b). Columns 3 and 4 use future sales growth ($SaleChange_t$) and value-added growth ($ValueAddChange_t$), respectively, as dependent variables to test the predictive power of a macroeconomic sentiment (i.e., *Economy*) score extracted from the text. The findings in these tables are highly consistent and further corroborate the pattern observed in the volatility and capital expenditure analysis. Both the $Economy_{RAW}$ score from the original text and the $Economy_{TRF}$ score from the anonymized text demonstrate strong predictive power for future sales and value-added growth, and the predictive coefficient of $Economy_{RAW}$ is robustly higher than that of $Economy_{TRF}$. This result once again confirms that anonymization leads to a degradation of informational value.

Finally, we assess the impact of anonymization on credit risk assessment by predicting future S&P credit rating downgrades. Donovan et al. (2021) demonstrate that machine learning methods can effectively extract qualitative credit risk indicators from earnings conference calls to predict future credit rating downgrades. We construct our own downgrade probability indicator utilizing GPT. The specific prompt used for this extraction is detailed in Appendix B.8. Column 5 reports the results of a horse-race regression where the dependent variable is an indicator for a rating downgrade within the following 12 months ($Downgrade_{t+1}$). We include the predicted downgrade probabilities derived from both raw ($\widehat{Downgrade}_{RAW}$) and anonymized ($\widehat{Downgrade}_{TRF}$) transcripts. Consistent with the results for volatility prediction, the coefficient on $\widehat{Downgrade}_{RAW}$ (0.037) is positive and statistically significant ($t = 3.31$). In contrast, the coefficient on $\widehat{Downgrade}_{TRF}$ is negative and statistically insignificant (-0.010, $t = -1.03$). This result implies that the raw text contains critical signals regarding credit deterioration that are effectively nullified by the anonymization process. When placed in competition, the signal from the anonymized text is subsumed by the raw text, indicating a substantial loss of predictive content relevant to credit risk.

Taken together, these results provide compelling evidence that the information loss from anonymization is a pervasive phenomenon that extends beyond sentiment analysis. The value of other types of information extracted from earnings call transcripts—ranging from volatility and investment signals to operating performance and credit risk indicators—is also compromised in this process, demonstrating the widespread nature of this degradation across various extraction tasks.

3.7 Alternative Textual Data

Our prior analyses focus on extracting information from conference call transcripts. In this subsection, we turn our attention to news headlines, another set of highly utilized textual data in finance setting (see e.g., Bybee et al., 2023; Lopez-Lira and Tang, 2023; Engelberg et al., 2025, for analyses using news or news headlines). Compared to the verbose nature of earnings call transcripts, news headlines are extremely short in length but feature a much higher density of information, particularly entity-related information.

We begin with a descriptive analysis of our news headline sample, with results presented in Table A.7. Our final sample comprises 52,716 news headlines from Yahoo Finance, corresponding to 30,950 firm-day observations (as a firm may have multiple news items on a given day), covering 2,052 unique firms over 292 trading days. The textual features of these headlines stand in stark contrast to those of earnings call transcripts. The average headline length is merely 15.55 tokens, with a highly concentrated distribution (a 25th percentile of 12 tokens and a 75th percentile of 18 tokens). Using the TRF model to anonymize all news headlines, we find that recognized entities account for an average of 38.30%, which is substantially higher than the 18.67% mean observed in earnings call transcripts under the same anonymization approach. This indicates that news headlines are highly condensed with entity-specific information, suggesting that anonymization could have an even more pronounced impact on their informational value. The detailed sample selection criteria for news headlines are provided in Panel B of Table A.1.

Following the methodology used for earnings call transcripts, we designed a specific prompt to extract sentiment from news headlines, which can be found in the Appendix

B.7. For each firm-day, we average the sentiment scores from all associated news headlines to derive a daily news sentiment score.

Table 9 presents the regression results for explaining contemporaneous stock returns using sentiment from news headlines. In both univariate (Columns 1 and 2) and multivariate regressions with standard controls (Columns 4 and 5), sentiment scores derived from raw ($Sentiment^{News}_{RAW}$) and TRF-anonymized ($Sentiment^{News}_{TRF}$) texts are significant positive signals of contemporaneous stock returns. This indicates that LLMs can extract market sentiment signals from short-form texts such as news headlines. Consistent with the findings from our analysis of earnings call transcripts, the regression coefficients and adjusted R^2 values are lower for the TRF-anonymized texts compared to the raw texts.

A key finding emerges from the "horse race" regressions (Columns 3 and 6), which include sentiment scores from both raw and anonymized texts in the same model specification. The results show clear evidence of information loss from anonymization. In the model without control variables (Column 3), the coefficient on $Sentiment^{News}_{RAW}$ (0.315) remains statistically significant, whereas the coefficient on $Sentiment^{News}_{TRF}$ (0.135) is attenuated to less than one-third of its magnitude in the standalone regression. This pattern persists after incorporating control variables (Column 6), where the coefficient on $Sentiment^{News}_{RAW}$ (0.314) remains substantially larger than that on $Sentiment^{News}_{TRF}$ (0.131).

Taken together, the results from the news headline analysis corroborate the findings from the earnings call transcripts. This suggests that the loss of information from anonymization is not an idiosyncratic phenomenon confined to a specific type of financial text. Whether the source is a lengthy earnings call transcript or a concise news headline, the named entities within the text are a significant source of its explanatory and predictive value. Consequently, removing these entities to mitigate potential look-ahead bias results in a corresponding loss of the text's intrinsic informational content.

3.8 Pre- and Post-Cutoff Date Analysis

So far, we have shown evidence that indicates anonymization can cause loss of information in LLM-extracted signals. An important implication from our study is that the extent to which anonymization methods can alleviate look-ahead bias needs to be assessed using both pre- and post-knowledge cutoff samples. Assessments based solely on the pre-knowledge-cutoff sample can fail to distinguish between the intended reduction in look-ahead bias and the unintended information loss caused by removing context.

To demonstrate our point, we next extend the earnings call transcript sample to include both the pre- and post-cutoff periods.[12] In this subsection, we extend our analysis to include the pre-cutoff period (November 2022–October 2023), allowing us to assess the impact of anonymization in a setting that is closer to prior research that uses anonymization to mitigate look-ahead bias. To ensure comparability between the pre- and post-cutoff periods, we retain only firms that appear in both samples. Throughout this section, we use GPT-4o-mini for information extraction and identification testing to generate the main results.

If look-ahead bias significantly inflates raw sentiment signals in the pre-cutoff period, the reduction in informativeness observed between raw and anonymized signals will be more pronounced in the pre-cutoff period compared to the post-cutoff period. This larger reduction would reflect both the inherent information loss from anonymization and the additional mitigation of look-ahead bias.

We first visually present the variation in the relative performance of the raw and anonymized signals over time. We conduct quarter-by-quarter regressions following the specification of Table 4. Figure 2 plots the quarterly time series of coefficients on the sentiment score. Panel A compares the coefficients derived from raw transcripts against those from TRF-anonymized samples, while Panel B compares raw against LLM-anonymized signals. All independent variables are winsorized and standardized within each quarter, and specifications are estimated quarter by quarter so that the coefficients are comparable over time.

[12] All LLMs used in this study share a knowledge cutoff of October 2023.

The knowledge cutoff (2023 Q4) is marked with a dashed vertical line. In both subfigures, the coefficient of $Sentiment_{RAW}$ (in dark blue) consistently exceeds that of the anonymized counterparts ($Sentiment_{TRF}$ and $Sentiment_{LLM}$, in red), indicating persistent and economically meaningful information loss in both the pre- and post-cutoff periods. The figure also reports the coefficient difference. The average magnitude of the difference is broadly similar on either side of the knowledge cutoff.

We then formally assess the impact of the knowledge cutoff and firm recognition on the informativeness of sentiment scores by estimating the following regression:

$$DGTW_{i,t} = \beta_1 Sentiment_{i,t} + \beta_2 (Sentiment_{i,t} \times Z_{i,t}) + \beta_3 Z_{i,t} + \gamma X_{i,t} + \delta_t + \epsilon_{i,t}, \quad (2)$$

This regression revises Regression (1) by introducing an interaction term between the sentiment signal and a proxy $Z_{i,t}$ for the effect of anonymization or look-ahead bias. The variable $Sentiment_{i,t}$ denotes either $Sentiment_{RAW}$, $Sentiment_{TRF}$, or $Sentiment_{LLM}$, and $X_{i,t}$ represents the same set of control variables as in Table 4.

To evaluate the impact of the knowledge cutoff and whether models can correctly identify firm information, we consider two interaction variables: pre-cutoff indicator *Pre*, which equals 1 if the transcript falls in the pre-knowledge-cutoff period (November 2022 to October 2023) and 0 otherwise, and the firm-identification indicator *Identified*, which equals 1 if the model correctly identifies the firm name or date in the anonymized transcript and 0 otherwise.

Table 10 implements specification (2) using GPT-4o-mini for sentiment extraction. We set $Z_{i,t} = Pre$ and use the full sample spanning both the pre- and post-cutoff periods. In column (1), we regress returns on $Sentiment_{RAW}$ and its interaction with Pre. The coefficient on $Sentiment_{RAW}$ is large and highly significant with a coefficient of 2.506 ($t = 19.564$), confirming that the raw text sentiment is strongly informative. The coefficient of the interaction term $Sentiment_{RAW} \times Pre$ is significantly negative ($t = -3.130$), implying that the predictive content of raw sentiment is somewhat weaker in the pre-cutoff period. Columns

(2) and (3) show a similar pattern for anonymized sentiment: the relationship between the extracted sentiment and stock returns is strongly positive for both $Sentiment_{TRF}$ (2.360, $t = 17.715$) and $Sentiment_{LLM}$ (2.397, $t = 17.913$), and the interaction terms with *Pre* are again negative and statistically significant (-0.488 and -0.515, respectively).

These results suggest that look-ahead bias does not play an important role in our sentiment extraction setting. If look-ahead bias strongly affected the extracted signals, one would expect the coefficients on the interaction terms to be positive and significant, reflecting artificially inflated predictive power in the pre-cutoff period when the LLM's training data include future outcomes. Instead, the significantly negative interactions indicate that sentiment, both raw and anonymized, is slightly less informative in the pre-cutoff window. This pattern is difficult to reconcile with a dominant role for look-ahead bias.

Next, we explicitly examine whether models' ability to identify firms from anonymized transcripts leads to significantly improved power in sentiment extraction. In Table 11, we interact extracted signals with *Identified* for pre- and post-knowledge cutoff samples. Across all specifications, while the main effects of anonymized sentiment remain highly significant, the interaction terms (*Sentiment* $\times$ *Identified*) are statistically insignificant. For instance, in the post-cutoff period for TRF (Column 2), the interaction is 0.069 ($t = 0.269$), and for LLM (Column 4), it is -0.371 ($t = -0.721$). The lack of a systematic relationship between identification and the informativeness of anonymized sentiment suggests that whether GPT-4o-mini can correctly identify the firm name is a poor proxy for the presence or severity of look-ahead bias or information loss from anonymization.[13]

Finally, we extend this analysis to a downgrade prediction task to examine the robustness

[13] Appendix Table A.8 explores the impact of anonymization using the *Gap* measure (the absolute difference between sentiment scores from RAW and anonymized transcripts) as $Z_{i,t}$ in specification (2). We consistently find that the interaction *Sentiment* $\times$ *Gap* is negative and highly significant across various anonymization methods (TRF and LLM) and sample periods (pre- and post-cutoff). These results indicate that larger discrepancies between raw and anonymized sentiment scores systematically correlate with weaker explanatory power of anonymized sentiment. Importantly, this result holds even in restricted subsamples (Columns 3 and 6) where we only include transcripts where the model failed to identify the firm name, effectively precluding look-ahead bias. Unlike the limited role of firm recognition, the *Gap* measure proves to be a reliable and economically meaningful proxy for anonymization-induced information loss.

of our findings in a different context. We replicate the analysis using S&P credit rating downgrades as the dependent variable. Figure A.1 and Tables A.9–A.10 in the Appendix present these results. The qualitative patterns closely mirror those in the sentiment setting: raw transcripts consistently yield higher predictive power than anonymized versions, the interaction with the pre-cutoff indicator is generally insignificant (suggesting no inflation from look-ahead bias), and the ability to identify the firm does not significantly recover predictive accuracy.

4 Conclusion

This study provides comprehensive empirical evidence that anonymization, a widely adopted technique to mitigate look-ahead bias in financial text analysis with LLMs, introduces a significant side effect: information loss. By systematically removing named entities, the process strips away essential context, thereby degrading the explanatory and predictive power of the extracted textual signals.

Our analysis, centered on earnings call transcripts, demonstrates that sentiment scores from raw text are strong signals of announcement-day stock returns. However, this explanatory ability is substantially attenuated after applying various anonymization methods. Horserace regressions that include both signals generated from the raw and anonymized transcripts confirm that the explanatory content of anonymized signals is largely subsumed by that of the raw text. We trace this degradation primarily to the removal of NUMBERS and AGENTS (e.g., firm and executive names), which serve as anchors for contextual interpretation. Furthermore, we establish that the divergence between raw and anonymized signals is more significant for transcripts with high linguistic uncertainty and a higher fraction of entity information, and firms with a poor information environment.

These findings are robust across multiple dimensions. The information loss persists across different anonymization models, various LLMs used for signal extraction (GPT-4o-mini, GPT-4o, and GPT-o3-mini), and extends beyond return explanation to forecasting future stock volatility, capital expenditures, and sales growth. Crucially, we replicate our core results using an alternative text corpus of short, entity-dense news headlines, confirming that the observed information loss is a fundamental consequence of the anonymization process itself, rather than an artifact of a specific textual domain.

Finally, we extend our analysis to the pre-cutoff period to assess the trade-off between bias mitigation and information loss under varying conditions. By examining two distinct prediction settings, we consistently observe pervasive and significant information loss. In both settings, the raw text significantly outperforms the anonymized text, even after accounting for potential look-ahead bias. This finding underscores that the degradation of signal quality is a fundamental consequence of anonymization that persists regardless of the severity of the bias it aims to mitigate. Additionally, our results caution that one should not assess the performance of anonymization technique in reducing look-ahead bias purely based on pre-knowledge cutoff sample, as it could confound the reduction in information in the transcript with the reduction in the look-ahead bias.

Our research challenges the default reliance on full anonymization in financial applications of LLMs. It highlights an important trade-off between mitigating look-ahead bias and preserving the intrinsic informational content of text. Future research should focus on developing methodologies that can effectively balance these competing objectives, ensuring that efforts to achieve methodological rigor do not inadvertently discard valuable information.

References

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The journal of finance*, 23(4):589–609.

Bernard, V. L. and Thomas, J. K. (1989). Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*.

Breitung, C. and Müller, S. (2025). Global business networks. *Journal of Financial Economics*.

Brown, S., Hillegeist, S. A., and Lo, K. (2004). Conference calls and information asymmetry. *Journal of Accounting and Economics*, 37(3):343–366.

Bybee, L., Kelly, B., and Su, Y. (2023). Narrative asset pricing: Interpretable systematic risk factors from news text. *The Review of Financial Studies*, 36(12):4759–4787.

Chen, S., Peng, L., and Zhou, D. (2025). Wisdom or whims? decoding the language of retail trading with social media and ai. *Asian Bureau of Finance and Economic Research*.

Chen, Y., Kelly, B. T., and Xiu, D. (2022). Expected returns and large language models. *Available at SSRN 4416687*.

Clayton, C., Coppola, A., Maggiori, M., and Schreger, J. (2025). Geoeconomic pressure. *Columbia Business School Research Paper Forthcoming, Stanford University Graduate School of Business Research Paper Forthcoming*.

Daniel, K., Grinblatt, M., Titman, S., and Wermers, R. (1997). Measuring mutual fund performance with characteristic-based benchmarks. *The Journal of Finance*.

Donovan, J., Jennings, J., Koharki, K., and Lee, J. (2021). Measuring credit risk using qualitative disclosure. *Review of Accounting Studies*, 26(2):815–863.

Engelberg, J., Manela, A., Mullins, W., and Vulicevic, L. (2025). Entity neutering. *Available at SSRN.*

Frankel, R., Johnson, M., and Skinner, D. J. (1999). An empirical examination of conference calls as a voluntary disclosure medium. *Journal of Accounting Research*, 37(1):133–150.

Garcia, D., Hu, X., and Rohrer, M. (2023). The colour of finance words. *Journal of Financial Economics.*

Glasserman, P. and Lin, C. (2023). Assessing look-ahead bias in stock return predictions generated by gpt sentiment analysis. *arXiv preprint arXiv:2309.17322.*

He, J. J. and Tian, X. (2013). The dark side of analyst coverage: The case of innovation. *Journal of financial economics*, 109(3):856–878.

He, S., Lv, L., Manela, A., and Wu, J. (2025). Chronologically consistent large language models. *arXiv preprint arXiv:2502.21206.*

Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *The Journal of Finance.*

Hoberg, G. and Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of political economy*, 124(5):1423–1465.

Jha, M., Qian, J., Weber, M., and Yang, B. (2024a). Chatgpt and corporate policies. *National Bureau of Economic Research.*

Jha, M., Qian, J., Weber, M., and Yang, B. (2024b). Harnessing generative ai for economic insights. *arXiv preprint arXiv:2410.03897.*

Ke, Z. T., Kelly, B. T., and Xiu, D. (2019). Predicting returns with text data. *National Bureau of Economic Research.*

Kim, A., Muhn, M., and Nikolaev, V. (2023). From transcripts to insights: Uncovering corporate risks using generative ai. *arXiv preprint arXiv:2310.17721*.

Levy, B. (2024). Caution ahead: Numerical reasoning and look-ahead bias in ai models. *Available at SSRN 5082861*.

Lopez-Lira, A. and Tang, Y. (2023). Can chatgpt forecast stock price movements? return predictability and large language models. *arXiv preprint arXiv:2304.07619*.

Lopez-Lira, A., Tang, Y., and Zhu, M. (2025). The memorization problem: Can we trust llms' economic forecasts? *arXiv preprint arXiv:2504.14765*.

Loughran, T. and Mcdonald, B. (2011). When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of Finance*.

Loughran, T. and Mcdonald, B. (2014). Measuring readability in financial disclosures. *The Journal of Finance*.

Ludwig, J., Mullainathan, S., and Rambachan, A. (2025). Large language models: An applied econometric framework. *National Bureau of Economic Research*.

Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of accounting research*, pages 109–131.

Sarkar, S. K. (2024). Storieslm: A family of language models with time-indexed training data. *Available at SSRN 4881024*.

Sarkar, S. K. (2025). Economic representations.

Sarkar, S. K. and Vafa, K. (2024). Lookahead bias in pretrained language models. *Available at SSRN 4754678*.

Siano, F. (2025). The news in earnings announcement disclosures: Capturing word context using llm methods. *Management Science*.

Tetlock, P. C. (2010). Does public financial news resolve asymmetric information? *The Review of Financial Studies*, 23(9):3520–3557.

Tetlock, P. C., Saar-Tsechansky, M., and Macskassy, S. (2008). More than words: Quantifying language to measure firms' fundamentals. *The Journal of Finance*.

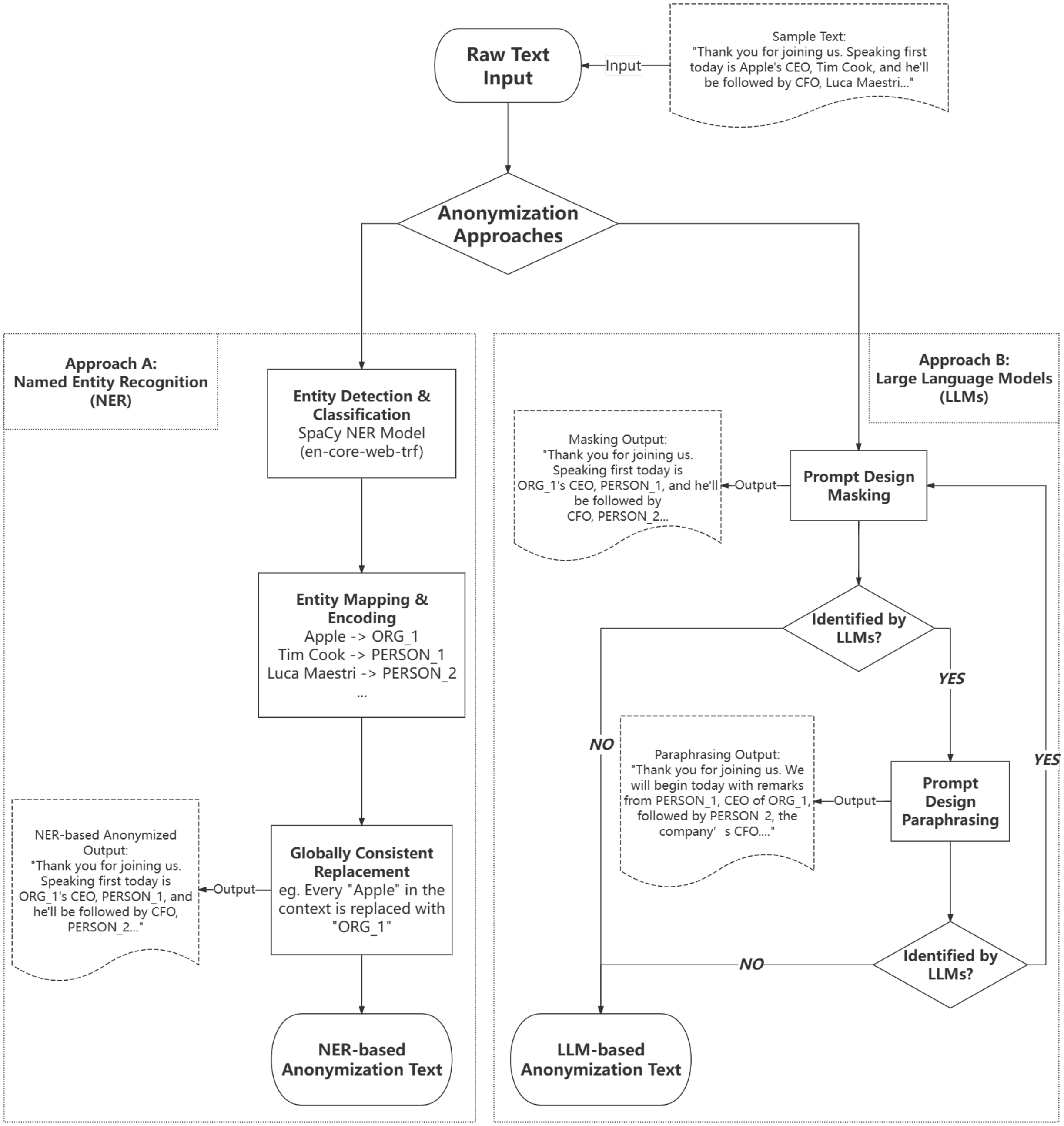
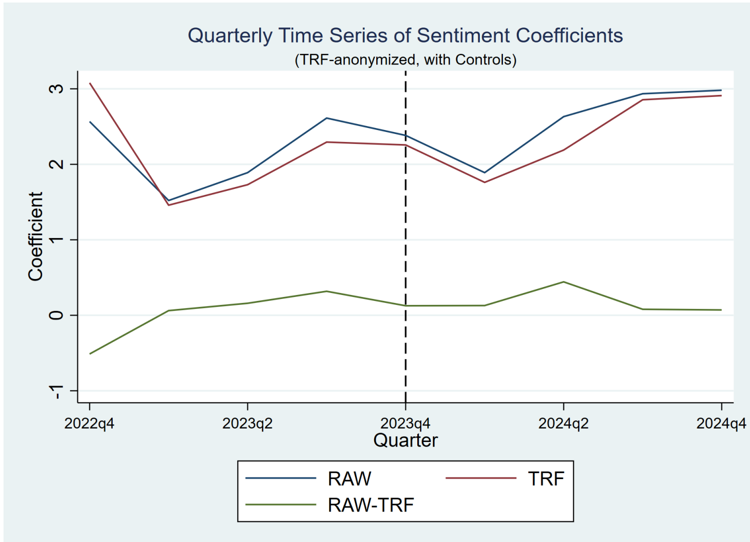
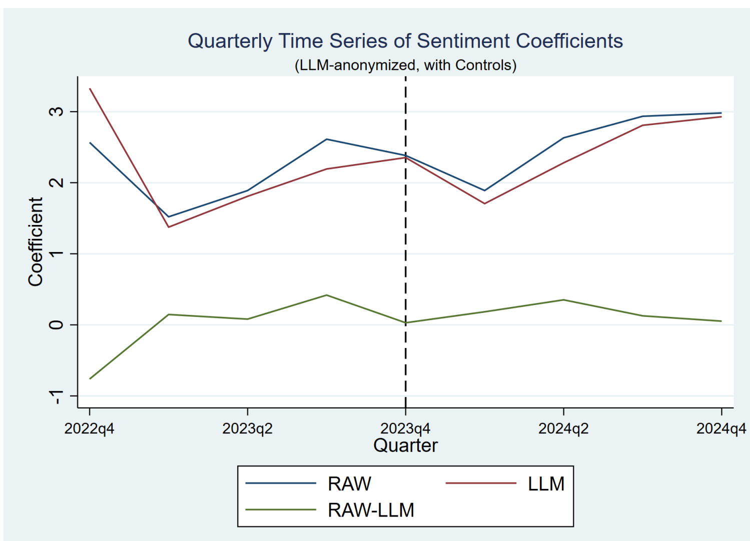


Figure 1: **Anonymization Pipeline**

This figure shows two text anonymization approaches applied to a sample from an Apple Inc. earnings call transcript. Approach A shows a Named Entity Recognition (NER) approach using spaCy model (TRF). This method employs a global mapping strategy, where each unique entity is consistently replaced by a numbered placeholder (e.g., PERSON_1, ORG_2) according to its order of appearance to maintain readability. Approach B utilizes a Large Language Model (LLM), specifically GPT-4o-mini, to perform anonymization through carefully designed prompts and an iterative process.



(a) Quarterly Sentiment Coefficients
(RAW and TRF-anonymized)



(b) Quarterly Sentiment Coefficients
(RAW and LLM-anonymized)

Figure 2: **Quarterly Time Series of Sentiment Coefficients**

This figure plots the quarterly time series of coefficients on sentiment scores, estimated separately from regressions with controls following the specification in Table 4. Sentiment scores are extracted using GPT-4o-mini from different versions of earnings call transcripts. Subfigure 2a compares the coefficients derived from the RAW transcripts against those from the TRF-anonymized samples. Subfigure 2b compares the coefficients from the RAW transcripts against those from the LLM-anonymized samples.

Table 1: **Comparison of Text Anonymization Approaches**

This table shows a comparative analysis of anonymization outputs, displaying the original raw text alongside anonymized versions processed by two distinct approaches: TRF and LLM (GPT-4o-mini). Replaced entities in the anonymized text are highlighted in bold for clarity.

Raw Text	**TRF Anonymized**	**LLM Anonymized**
Please note that some of the information you'll hear during our discussion today will consist of forward-looking statements, including, without limitation, those regarding revenue, gross margin, operating expenses, other income and expense, taxes, capital allocation and future business outlook, including the potential impact of macroeconomic conditions on the firm's business and results of operations. These statements involve risks and uncertainties that may cause actual results or trends to differ materially from our forecast. For more information, please refer to the risk factors discussed in Apple's most recently filed annual report on Form 10-K and the form 8-K filed with the SEC today along with the associated press release. Apple assumes no obligation to update any forward-looking statements, which speak only as of the date they are made.	Please note that some of the information you'll hear during our discussion **DATE_1** will consist of forward-looking statements, including, without limitation, those regarding revenue, gross margin, operating expenses, other income and expense, taxes, capital allocation and future business outlook, including the potential impact of macroeconomic conditions on the firm's business and results of operations. These statements involve risks and uncertainties that may cause actual results or trends to differ materially from our forecast. For more information, please refer to the risk factors discussed in **ORG_1**'s most recently filed **DATE_2** report on Form 10-K and the form 8-K filed with the **ORG_2 DATE_1** along with the associated press release. **ORG_1** assumes no obligation to update any forward-looking statements, which speak only as of the date they are made.	Please be aware that some of the information shared during our conversation today will include forward-looking statements, which may encompass, but are not limited to, projections regarding revenue, gross margin, operating expenses, other income and expenses, taxes, capital allocation, and future business outlook, including potential effects of macroeconomic conditions on the entity's operations and results. These statements carry risks and uncertainties that could lead to actual outcomes differing significantly from our predictions. For further details, please refer to the risk factors outlined in **Entity˙A**'s most recent annual report on Form 10-K and the Form 8-K submitted to the regulatory body today, along with the related press release. **Entity˙A** has no obligation to update any forward-looking statements, which are valid only as of the date they are made.

Table 2: **Conference Call Entity Count and Anonymization Performance**

This table presents the summary statistics for conference call transcripts. In Panel A, we report the mean, standard deviation, and percentile distributions of earnings call transcript lengths and entity percentages identified by two methods: TRF and LLM (GPT-4o-mini). Lengths are measured in number of tokens. Entity% is calculated using the anonymized transcript, where the percentage represents the ratio of anonymized entity tokens to the total tokens in the anonymized version. In Panel B, we report the re-identification rates for four key attributes: Date, Firm Name, Ticker, and Industry under two different anonymization approaches. We define a correct match using the following criteria: the estimated Date must fall within 7 days of the actual conference call date; Firm Name and Industry are evaluated using fuzzy string matching (Levenshtein token sort ratio), where a score of 75 or higher after text normalization is considered a match; and Ticker symbols require an exact case-insensitive match with the ground truth.

Panel A: Transcript Length and Entity Percentages					
	Mean	**Std. Dev.**	**25th Pct.**	**Median**	**75th Pct.**
Length	8,895	2,532	6,971	9,016	10,901
Entity% (TRF)	18.67	2.71	16.84	18.52	20.39
Entity% (LLM)	11.18	2.97	9.23	10.95	12.84

Panel B: Identification Ratio %		
Date	5.39	3.09
Firm Name	62.71	5.49
Ticker	50.26	5.54
Industry	52.75	49.65
Anonymization Approach	TRF	GPT-4o-mini
Entity Remove	ALL	ALL
Recognition Approach	GPT-4o-mini	GPT-4o-mini

Table 3: **Correlations of Sentiment Signals**

This table shows correlations among sentiment scores extracted from raw transcripts (RAW) and anonymized transcripts from two different anonymization approaches (TRF and LLM). All sentiment measures are extracted using the GPT-4o-mini model. All values are Pearson correlation coefficients.

	$Sentiment_{RAW}$	$Sentiment_{TRF}$	$Sentiment_{LLM}$
$Sentiment_{RAW}$	1.000		
$Sentiment_{TRF}$	0.864	1.000	
$Sentiment_{LLM}$	0.858	0.859	1.000

Table 4: **Explanatory Power Comparison of Sentiment Scores with controls**

This table presents the results of regressions with DGTW-adjusted returns on the earnings conference call day as dependent variables. The key independent variables are sentiment measures extracted using GPT-4o-mini from earnings conference call transcripts. Our control variables including standard drivers of stock returns, including analyst forecast error (FE), past returns at various horizons, and firm characteristics such as size, book-to-market ratio, and turnover. Columns (1) through (3) augment this baseline model by individually adding the sentiment scores derived from the four text sources. Columns (4) and (5) include both the sentiment extracted from raw transcript and from an anonymized transcript. All variables are winsorized at the 1st and 99th percentiles. All independent variables are standardized. All regressions include date fixed effects. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$				
VARIABLES	(1)	(2)	(3)	(4)	(5)
$Sentiment_{RAW}$	2.487***			1.822***	1.808***
	(18.832)			(8.475)	(7.234)
$Sentiment_{TRF}$		2.331***		0.775***	
		(17.287)		(3.662)	
$Sentiment_{LLM}$			2.329***		0.798***
			(17.466)		(3.247)
FE	2.054***	2.073***	2.077***	2.048***	2.048***
	(12.622)	(12.595)	(12.778)	(12.608)	(12.660)
$DGTW_{i,t-1}$	-0.335***	-0.345***	-0.337***	-0.344***	-0.341***
	(-2.838)	(-2.877)	(-2.831)	(-2.901)	(-2.892)
$DGTW_{i,t-2}$	-0.050	-0.009	-0.023	-0.043	-0.048
	(-0.350)	(-0.060)	(-0.157)	(-0.299)	(-0.333)
$DGTW_{i,t-3}$	0.132	0.165	0.143	0.135	0.127
	(1.039)	(1.301)	(1.114)	(1.069)	(1.002)
$DGTW_{i,t-22,t-4}$	-0.419***	-0.393***	-0.407***	-0.425***	-0.430***
	(-3.107)	(-2.875)	(-3.031)	(-3.140)	(-3.199)
$DGTW_{i,t-253,t-23}$	-0.331**	-0.290**	-0.271*	-0.347**	-0.342**
	(-2.348)	(-2.120)	(-1.969)	(-2.494)	(-2.449)
$ln(Size)$	-0.434***	-0.364***	-0.388***	-0.422***	-0.431***
	(-3.673)	(-3.156)	(-3.358)	(-3.574)	(-3.670)
$ln(BM)$	-0.194*	-0.194*	-0.177	-0.191	-0.185
	(-1.657)	(-1.656)	(-1.514)	(-1.633)	(-1.580)
$ln(Turnover)$	0.201*	0.196*	0.171	0.201*	0.192*
	(1.810)	(1.746)	(1.509)	(1.807)	(1.728)
Date FE	Yes	Yes	Yes	Yes	Yes
Observations	7352	7352	7352	7352	7352
Adj. R^2	0.132	0.124	0.124	0.133	0.133

Table 5: **Comparing Explanatory Power Under Anonymizing Different Categories of Entities**

This table investigates the source of information loss by examining the impact of anonymizing specific entity categories. The selective replacement of these entities is performed using the TRF model, based on the definitions for each category provided in Appendix Table A.4. The dependent variable is the DGTW-adjusted stock return ($DGTW_{i,t}$). Columns (1) through (5) present results for sentiment scores derived from texts where only one category of entities—NUMBERS, PLACES, AGENTS, or OTHERS—has been replaced, benchmarked against the original RAW text score. Columns (6) through (9) conduct "horse-race" regressions, directly comparing the explanatory power of the sentiment from the original RAW text against the sentiment from each of the selectively anonymized texts. Control variables are identical to those in Table 4. All variables are winsorized at the 1st and 99th percentiles. All independent variables are standardized. All regressions include date fixed effects. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$								
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Sentiment_{RAW}$	2.487***					1.845***	1.397***	1.767***	1.009***
	(18.832)					(7.998)	(4.499)	(6.317)	(2.666)
$Sentiment_{NUM}$		2.336***				0.737***			
		(16.934)				(3.155)			
$Sentiment_{PLC}$			2.476***				1.158***		
			(18.194)				(3.687)		
$Sentiment_{AGT}$				2.397***				0.795***	
				(18.375)				(2.993)	
$Sentiment_{OTH}$					2.527***				1.583***
					(20.124)				(4.442)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7352	7352	7352	7352	7352	7352	7352	7352	7352
Adj.R^2	0.132	0.124	0.131	0.127	0.134	0.133	0.133	0.133	0.135

Table 6: **Determinants of Sentiment Score Divergence**

This table examines the factors influencing the absolute difference in sentiment scores between RAW and TRF-anonymized transcripts ($Gap_{i,t} = |Sentiment_{RAW} - Sentiment_{TRF}|$). Each column presents a univariate regression of this gap on specific textual or firm-level characteristics. Column (1) analyzes the impact of textual uncertainty in the original transcript. Columns (2) through (5) investigate the role of the firm's information environment, including firm size, analyst coverage, product similarity (TNIC3TSIMM), and the proportion of entities identified by the TRF model. All independent variables are standardized, and all variables are winsorized at the 1st and 99th percentiles. Date fixed effects are included across all specifications. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $Gap_{i,t}$				
VARIABLES	(1)	(2)	(3)	(4)	(5)
$Uncertainty_{RAW}$	0.040***				
	(12.080)				
$ln(Size)$		-0.008***			
		(-3.337)			
$Coverage$			-0.005**		
			(-2.003)		
$TNIC3TSIMM$				-0.006***	
				(-3.416)	
$Entity\%$ (TRF)					0.004*
					(1.682)
Date FE	Yes	Yes	Yes	Yes	Yes
Observations	7352	7352	7352	6860	7352
Adj.R^2	0.042	0.005	0.004	0.005	0.004

Table 7: **Information Loss of Alternative LLMs**

This table compares sentiment scores generated by three distinct models: GPT-4o-mini, GPT-4o, and GPT-o3-mini. All regressions include the full set of control variables and use the DGTW-adjusted stock return ($DGTW_{i,t}$) as the dependent variable. Columns (1) through (3) report the results for sentiment extracted from the raw transcripts by each LLM. Columns (4) through (6) report the results for sentiment extracted from the TRF-anonymized text. Finally, Columns (7) through (9) conduct "horse-race" regressions for each LLM-extracted signals. Control variables are identical to Table 4. All variables are winsorized at the 1st and 99th percentiles. All independent variables are standardized. All regressions include date fixed effects. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$								
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Sentiment_{RAW}$	2.487***	2.528***	2.400***				1.822***	2.105***	1.660***
	(18.832)	(19.972)	(18.968)				(8.475)	(13.157)	(9.504)
$Sentiment_{TRF}$				2.331***	2.005***	2.229***	0.775***	0.611***	0.942***
				(17.287)	(16.879)	(16.412)	(3.662)	(4.337)	(5.069)
LLM	4o-mini	4o	o3-mini	4o-mini	4o	o3-mini	4o-mini	4o	o3-mini
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7352	7352	7352	7352	7352	7352	7352	7352	7352
Adj.R^2	0.132	0.131	0.125	0.124	0.108	0.118	0.133	0.133	0.129

Table 8: **Information Loss in Alternative Tasks**

This table compares the predictive power of textual scores extracted from RAW and TRF-anonymized earnings-call transcripts in forecasting five forward-looking outcomes. Each column reports the results from a "horse-race" regression where both the RAW and TRF-anonymized scores are included simultaneously. Column (1) examines realized post-announcement stock return volatility (Vol_{post}) using managerial uncertainty ($Uncertainty_{RAW}$ and $Uncertainty_{TRF}$), controlling for pre-announcement volatility (Vol_{pre}), pre-announcement alpha ($Alpha_{pre}$), absolute abnormal return ($|Abnormal\ Ret|$), firm size (ln($Size$)), and book-to-market ratio (ln(BM)). Column (2) predicts firms' capital expenditures two quarters ahead ($Capx_{t+2}$) using investment expectations ($Investment_{RAW}$ and $Investment_{TRF}$), controlling for current capital expenditure ($Capx_t$), financial leverage ($Leverage_{Market}$), and firm size (ln($Total\ Asset$)). Column (3) investigates two-quarter sales growth ($SaleChange_t$) with firms' expressed outlook on the economy ($Economy_{RAW}$ and $Economy_{TRF}$), controlling for lagged two-quarter sales growth ($SaleChange_{t-2}$), book-to-market ratio (BM), firm size (ln($Total\ Asset$)), and asset tangibility ($Tangibility$). Column (4) presents analogous predictive regression of two-quarter value-added growth ($ValueAddChange_t$), using the same set of control variables as in Column (3). Column (5) tests the predictive power for future S&P credit rating downgrades ($Downgrade_{t+1}$), where the dependent variable is an indicator equaling 1 if the firm is downgraded within 12 months. The key predictors are the predicted downgrade probabilities extracted from raw ($\widehat{Downgrade}_{RAW}$) and TRF-anonymized ($\widehat{Downgrade}_{TRF}$) texts. Control variables for Column (5) include $Leverage_{Asset}$, MTB, ROA, $RevGrowth$, $Std(Income)$, $Std(Revenue)$, $Zscore$ and $Oscore$. All variable detailed definitions are listed in Table A.2. All regressions include fixed effects as indicated. All variables are winsorized at the 1st and 99th percentiles. All independent variables are standardized. T-statistics based on time-clustered (Cols 1-4) or SIC3-clustered (Col 5) standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Vol_{post}	$Capx_{t+2}$	$SaleChange_t$	$ValueAddChange_t$	$Downgrade_{t+1}$
VARIABLES	(1)	(2)	(3)	(4)	(5)
$Uncertainty_{RAW}$	0.026**				
	(2.028)				
$Uncertainty_{TRF}$	0.012				
	(0.777)				
$Investment_{RAW}$		0.043***			
		(3.482)			
$Investment_{TRF}$		0.033**			
		(2.502)			
$Economy_{RAW}$			1.633***	2.889***	
			(4.207)	(5.922)	
$Economy_{TRF}$			1.380***	1.409***	
			(3.451)	(3.150)	
$\widehat{Downgrade}_{RAW}$					0.037***
					(3.31)
$\widehat{Downgrade}_{TRF}$					-0.010
					(-1.03)
Controls	Yes	Yes	Yes	Yes	Yes
Fixed Effect	Date	Date	Date	Date	SIC3 × Quarter
Observations	7255	5692	6630	6365	4352
Adj.R^2	0.776	0.730	0.212	0.249	0.149

Table 9: **Anonymization and Information Loss from News Headlines**

This table examines if information loss also affects news headlines. The dependent variable is DGTW-adjusted stock returns ($DGTW_{i,t}$) on news days. The sentiment scores are derived from RAW and TRF-anonymized headlines. Columns (1) and (2) show univariate regressions for the RAW and TRF-anonymized scores, respectively. Column (3) conducts a horse-race regression between the RAW and TRF-anonymized sentiment scores. Columns (4) through (6) replicate this analysis with control variables, which include past returns at various horizons, and firm characteristics such as size, book-to-market ratio, and turnover. All variables are winsorized at the 1st and 99th percentiles and standardized. All regressions include date fixed effects. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$					
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
$Sentiment^{News}_{RAW}$	0.444***		0.315***	0.439***		0.314***
	(17.477)		(4.664)	(17.353)		(4.633)
$Sentiment^{News}_{TRF}$		0.437***	0.135**		0.432***	0.131**
		(17.325)	(2.036)		(17.212)	(1.974)
$DGTW_{i,t-1}$				-0.036	-0.033	-0.036
				(-1.061)	(-0.981)	(-1.060)
$DGTW_{i,t-2}$				-0.041	-0.041	-0.041
				(-1.237)	(-1.254)	(-1.251)
$DGTW_{i,t-3}$				0.006	0.007	0.006
				(0.191)	(0.230)	(0.199)
$DGTW_{i,t-22,t-4}$				0.036	0.038	0.036
				(0.998)	(1.046)	(1.008)
$DGTW_{i,t-253,t-23}$				0.030	0.032	0.030
				(0.977)	(1.023)	(0.980)
$ln(Size)$				-0.075***	-0.074***	-0.073***
				(-2.984)	(-2.923)	(-2.910)
$ln(BM)$				-0.038*	-0.040*	-0.038*
				(-1.737)	(-1.805)	(-1.740)
$ln(Turnover)$				0.043	0.044	0.044
				(1.531)	(1.554)	(1.566)
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	30950	30950	30950	30950	30950	30950
Adj.R^2	0.016	0.015	0.016	0.016	0.016	0.016

Table 10: **Sample Period and Explanatory Power of Sentiment**

This table examines whether the explanatory power of sentiment scores extracted from raw and anonymized earnings call transcripts differ between pre- and post-knowledge cut-off date subsamples. The dependent variable is the contemporaneous DGTW-adjusted stock return ($DGTW_{i,t}$) on the earnings-announcement day. $Sentiment_{TRF}$ and $Sentiment_{LLM}$ denote sentiment scores computed from TRF-anonymized and LLM-anonymized transcripts, respectively. Pre is an indicator variable, which equals 1 if the observations belong to pre cut-off date subsample, and 0 for post subsample. We use sentiment score extracted from raw, TRF-anonymized, and LLM-anonymized transcripts, in three columns, respectively. All variables are winsorized at the 1st and 99th percentiles and standardized. All specifications include date fixed effects and control variables identical to previous tables. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

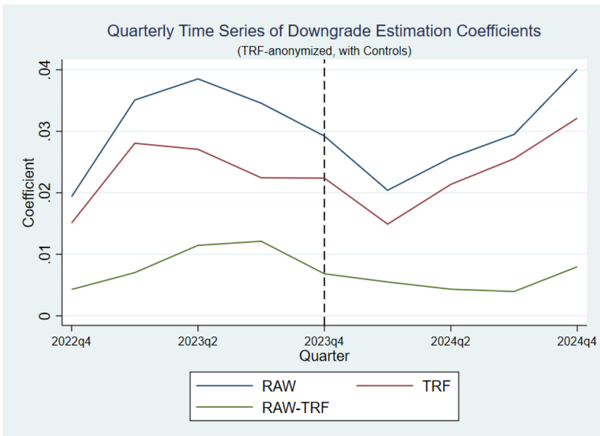
	Dependent Variable: $DGTW_{i,t}$		
VARIABLES	(1)	(2)	(3)
$Sentiment_{RAW} \times Pre$	-0.553***		
	(-3.130)		
$Sentiment_{RAW}$	2.506***		
	(19.564)		
$Sentiment_{TRF} \times Pre$		-0.488***	
		(-2.728)	
$Sentiment_{TRF}$		2.360***	
		(17.715)	
$Sentiment_{LLM} \times Pre$			-0.515***
			(-2.900)
$Sentiment_{LLM}$			2.397***
			(17.913)
Controls	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Observations	12,772	12,772	12,772
Adj.R^2	0.132	0.125	0.126

Table 11: **Identification and Explanatory Power of Sentiment**

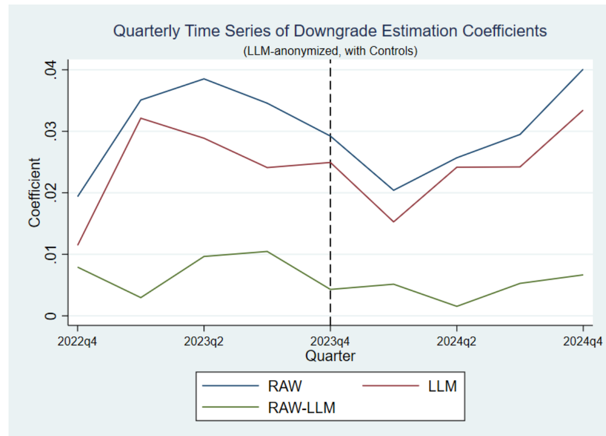
This table examines whether identification of anonymized earnings call transcripts affects the informativeness of extracted sentiment scores. The dependent variable is the contemporaneous DGTW-adjusted stock return ($DGTW_{i,t}$) on the earnings-announcement day. $Sentiment_{TRF}$ and $Sentiment_{LLM}$ denote sentiment scores computed from TRF-anonymized and LLM-anonymized transcripts, respectively. $Identified$ is an indicator variable, which equals 1 if GPT-4o-mini can identify the date or firm name of anonymized text, and equals 0 otherwise. Columns (1)–(2) use the TRF-anonymized sentiment scores, split into Pre and Post knowledge cutoff periods. Columns (3)–(4) use the LLM-anonymized sentiment scores for the same periods. All variables are winsorized at the 1st and 99th percentiles and standardized. All specifications include date fixed effects and control variables identical to previous tables. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$			
VARIABLES	(1)	(2)	(3)	(4)
$Sentiment_{TRF} \times Identified_{TRF}$	0.055	0.069		
	(0.202)	(0.269)		
$Identified_{TRF}$	0.325	0.440*		
	(1.337)	(1.883)		
$Sentiment_{TRF}$	1.850***	2.294***		
	(7.761)	(10.184)		
$Sentiment_{LLM} \times Identified_{LLM}$			-0.460	-0.371
			(-1.631)	(-0.721)
$Identified_{LLM}$			-0.465	-0.288
			(-1.398)	(-0.501)
$Sentiment_{LLM}$			1.963***	2.400***
			(14.438)	(16.514)
Period	Pre	Post	Pre	Post
Controls	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Observations	5,666	7,106	5,666	7,106
Adj. R^2	0.120	0.129	0.123	0.129

A Appendix



(a) Quarterly Downgrade Predictive Coefficients
(RAW and TRF-anonymized)



(b) Quarterly Downgrade Predictive Coefficients
(RAW and LLM-anonymized)

Figure A.1: **Quarterly Time Series of Downgrade Predictive Coefficients**

This figure plots the time series of the quarterly coefficients (β) on the predicted downgrade probability ($\widehat{Downgrade}_{i,t}$). The coefficients are estimated from cross-sectional regressions run separately for each calendar quarter, based on the following specification: $Downgrade_{i,t+1} = \alpha + \beta\widehat{Downgrade}_{i,t} + \Gamma\text{Controls}_{i,t} + \text{FE}_{SIC3} + \epsilon_{i,t}$ The model includes 3-digit SIC industry fixed effects (FE_{SIC3}), and standard errors are clustered at the SIC3 level. The set of control variables follows the specification in Table A.9. The predictive scores are extracted using GPT-4o-mini from different versions of earnings call transcripts. Subfigure A.1a compares the coefficients derived from the RAW transcripts against those from the TRF-anonymized samples. Subfigure A.1b compares the coefficients from the RAW transcripts against those from the LLM-anonymized samples. The horizontal dashed lines represent the average coefficients for the pre- and post-cutoff periods, and the vertical arrows indicate the gap in predictive power between the raw and anonymized versions.

Table A.1: **Sample Selection Criteria**

This table outlines the filtering procedure used to construct the final samples. Panel A details the selection process for earnings call transcripts, and Panel B for news headlines. Each row sequentially applies a filter, reporting the number of observations removed and the resulting sample size.

Panel A: Earnings Call Transcripts

Filter	**Sample Size**	**Obs. Removed**
Earnings call transcripts from Seeking Alpha for iShares Russell 3000 ETF constituents (Nov 2023 - Dec 2024)	10,812	
Remove duplicates, match dates and PERMNOs, remove errors	10,628	184
Remove transcripts with > 15,000 tokens	10,529	99
Match with DGTW returns (Nov 2023 - Dec 2024)	8,479	2,050
Match with analyst EPS forecast errors	7,596	883
Removing observations with missing controls	7,382	214

Panel B: News Headlines

Filter	**Sample Size**	**Obs. Removed**
Firm news headlines from Finnhub for iShares Russell 3000 ETF constituents (Nov 2023 - Dec 2024)	1,612,948	
Remove duplicates, require valid dates and non-missing headlines	969,077	643,871
Retain headlines from Yahoo, exclude "stock" or "stocks"	271,194	697,883
Retain headlines with a single organization mention	118,797	152,397
Remove weekend news and match with PERMNO	112,835	5,962
Aggregate by PERMNO and date	80,337	32,498
Match with DGTW returns (Nov 2023 - Dec 2024)	57,603	22,734
Removing observations with missing controls	30,950	26,653

Table A.2 **Variable Definitions**

This table defines the variables used in our analysis. Panel A lists measures derived from raw and anonymized earnings call transcripts and news headlines; all measures are extracted by LLMs through designed prompts listed in Appendix B.3 to B.7. Panel B lists the dependent variables. Panel C lists the control variables. Unless otherwise specified, stock return data are from CRSP, and firm-level accounting data are from Compustat.

Variable	**Definition**
Panel A: LLM Measures	
$\widehat{Downgrade}_{LLM}$	The predicted probability of an S&P credit rating downgrade within the 12 months following the earnings call, derived from LLM-anonymized earnings call transcripts.
$\widehat{Downgrade}_{RAW}$	The predicted probability of an S&P credit rating downgrade within the 12 months following the earnings call, derived from raw earnings call transcripts.
$\widehat{Downgrade}_{TRF}$	The predicted probability of an S&P credit rating downgrade within the 12 months following the earnings call, derived from TRF-anonymized earnings call transcripts.
$Economy_{RAW}$	Firm's view on macroeconomic trends of the original earnings call transcript. It is defined as a classification of the firm's anticipated change in optimism about the US economy over the next quarter. The score is categorized into one of five levels (Increase substantially, increase, no change, decrease, decrease substantially) based on the prompt derived from Jha et al. (2024b).
$Economy_{TRF}$	Firm's view on macroeconomic trends of the TRF-anonymized earnings call transcript.
$Investment_{RAW}$	Firm's investment score of the original earnings call transcript. It is defined as a classification of the firm's planned capital spending changes over the next year. The score is categorized into one of five levels (Increase substantially, increase, no change, decrease, decrease substantially) based on the prompt derived from Jha et al. (2024a).
$Investment_{TRF}$	Firm's investment score of the TRF-anonymized earnings call transcript.
$Sentiment_{AGT}$	Sentiment of the transcript where only AGENTS entity types are anonymized by the TRF model.
$Sentiment_{LLM}$	Sentiment of the earnings call transcript text after anonymization using GPT-4o-mini.
$Sentiment_{NUM}$	Sentiment of the transcript where only NUMBERS entity types are anonymized by the TRF model.
$Sentiment_{OTH}$	Sentiment of the transcript where only OTHERS entity types are anonymized by the TRF model.

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Table A.2 – Continued from previous page

Variable	Definition
$Sentiment_{PLC}$	Sentiment of the transcript where only PLACES entity types are anonymized by the TRF model.
$Sentiment_{RAW}$	Sentiment of the original earnings call transcript text. It is defined as a measure of its positive or negative implication for stock returns. The sentiment score, which ranges from -1 to 1, is computed with the formula Sentiment = (2 * Direction - 1) * Magnitude, where Direction captures the binary tone and Magnitude captures its intensity.
$Sentiment_{RAW}^{News}$	Sentiment of original Yahoo news headlines, which is defined as a measure of its positive or negative implication for stock returns. The sentiment score, which ranges from -1 to 1, is computed with the formula Sentiment = (2 * Direction - 1) * Magnitude, where Direction captures the binary tone and Magnitude captures its intensity.
$Sentiment_{TRF}$	Sentiment of the earnings call transcript text after anonymization using the spaCy en-core-web-trf (TRF) model.
$Sentiment_{TRF}^{News}$	Sentiment of the TRF-anonymized Yahoo news headlines.
$Uncertainty_{RAW}$	Uncertainty score of the original earnings call transcript. It is defined as a measure of the level of imprecision and cautious tone conveyed in the text. The score, which ranges from 0 (absolute certainty) to 1 (maximum uncertainty), is extracted based on the prompt derived from Loughran and Mcdonald (2011).
$Uncertainty_{TRF}$	Uncertainty score of the TRF-anonymized earnings call transcript.
Panel B: Dependent Variables	
$Capx_{t+2}$	Capital expenditure (CAPX) at the end of quarter $t+2$, scaled by total assets.
$DGTW_{i,t}$	DGTW-adjusted stock return on the announcement date (day 0), defined as Daniel et al. (1997).
$Downgrade_{i,t+1}$	An indicator variable equaling 1 if S&P downgrades the firm's credit rating within the following 12 months from the earnings call release, and 0 otherwise.
$Gap_{i,t}$	Absolute difference between the sentiment of original and TRF-anonymized transcripts.
$SaleChange_t$	Defined as $\log(Sale_{t+1}/Sale_{t-1})$; represents two-quarter sales growth.
$ValueAddChange_t$	Defined as $\log(ValueAdd_{t+1}/ValueAdd_{t-1})$; represents two-quarter value-added growth.
Vol_{post}	Standard deviation of stock returns over the post-announcement window [0, 22], requiring at least 10 daily observations.
Panel C: Control Variables	

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Table A.2 – Continued from previous page

Variable	Definition
|Abnormal Ret|	Absolute abnormal return, measured as the two-day (day 0 to +1) buy-and-hold return minus the buy-and-hold return of the CRSP value-weighted index.
$Alpha_{pre}$	Alpha from a market model estimated over the window [-252, -6], requiring at least 60 daily observations.
BM	Book-to-market ratio at the end of June, calculated as book equity (fiscal year-end) divided by market equity (fiscal year-end).
$Capx_t$	Capital expenditure (CAPX) at the end of quarter t, scaled by total assets.
$Coverage$	The average of the 12 monthly numbers of earnings forecasts from I/B/E/S, defined as He and Tian (2013).
$DGTW_{i,t-1}$	DGTW-adjusted stock return on event day -1.
$DGTW_{i,t-2}$	DGTW-adjusted stock return on event day -2.
$DGTW_{i,t-3}$	DGTW-adjusted stock return on event day -3.
$DGTW_{i,t-22,t-4}$	DGTW-adjusted stock return over the event window [-22, -4].
$DGTW_{i,t-253,t-23}$	DGTW-adjusted stock return over the event window [-253, -23].
FE	Analyst forecast error, defined as Hirshleifer et al. (2009).
$Leverage_{Asset}$	Total debt scaled by total assets ((dlcq + dlttq)/atq).
$Leverage_{Market}$	Financial leverage, calculated as total debt (dlttq + dlcq) divided by total debt plus market value of equity.
MTB	Market value of equity scaled by book value of equity ((prccq * cshoq) / ceqq).
$Oscore$	O-Score measure of credit risk following Ohlson (1980).
$RevGrowth$	percentage changes in revenue (revtq) from the same quarter, prior year.
ROA	Income before extraordinary items scaled by total assets (ibq / atq).
$Size$	Market capitalization at the end of June, calculated as CRSP price times shares outstanding.
$Std(Income)$	Standard deviation of quarterly income before extraordinary items measured over the previous twenty quarters.
$Std(Revenue)$	Standard deviation of total revenues measured over the previous twenty quarters.
$Tangibility$	Plant, Property, and Equipment (PPENTQ) scaled by total assets (atq) at the end of the quarter.
$TNIC3TSIMM$	A firm-specific measure of total product similarity against rivals in 2023 based on the Text-Based Network Industry Classification (TNIC), defined as Hoberg and Phillips (2016).
$Total\ Asset$	Logarithm of quarterly total assets from Compustat (atq).
$Turnover$	Share turnover at the end of June, calculated as prior year's trading volume divided by shares outstanding.
Vol_{pre}	Standard deviation of stock returns over the pre-announcement window [-252, -1], requiring at least 60 daily observations.

Continued on next page

Table A.2 – Continued from previous page

Variable	**Definition**
$Zscore$	Z-Score measure of bankruptcy risk following Altman (1968).

Table A.3: **Summary Statistics of Earnings Call Transcripts Sample**

This table presents summary statistics for earnings call transcripts from November 2023 through December 2024 in our sample. Panel A provides an overview of the sample, including the number of earnings call transcripts, firms, and dates, as well as the percentage of earnings calls held before market close and the percentage of firms listed on NASDAQ. Panel B presents detailed distributional statistics for firm-level fundamental variables, such as firm size (market capitalization), book-to-market ratio, and stock turnover. Firm size is reported in millions of dollars.

Panel A: Sample Overview

Transcripts	Firms	Dates	Before Mkt Close (%)	NASDAQ (%)
7,382	1,831	259	64.21	47.41

Panel B: Fundamental Statistics

	Size (Million $)	Book-to-Market	Turnover
Mean	23,845	0.55	2.77
Std. Dev.	137,878	0.52	8.04
25th Percentile	1,119	0.23	1.37
Median	3,293	0.44	1.92
75th Percentile	10,632	0.75	2.91

Table A.4: **Entity Category**

This table presents the 18 mutually exclusive entity types identified by the en-core-web-trf NER model from Python's spaCy library. These entity types are categorized into four primary groups: NUMBERS, PLACES, AGENTS, and OTHERS. For each entity type, the table includes its description, an example from an earnings call transcript, the total frequency of occurrences in the transcript corpus, and the average frequency of each of the four main categories per transcript. The last two columns show the total frequency of each entity type in news headlines, along with the average frequency of each of the four main categories per headline.

Category	Entity Type	Description	Example	Total Frequency (Transcript)	Frequency per Transcript	Total Frequency (Headlines)	Frequency per Headline
NUMBERS	DATE	Absolute or relative dates or periods	Q1 2024	1247244	300.72	21862	0.678
	CARDINAL	Numerals that do not fall under another type	one	274350		4670	
	MONEY	Monetary values, including unit	$69.7 billion	261834		5419	
	PERCENT	Percentage, including "%"	6%	250714		2322	
	ORDINAL	"first", "second", etc.	first	92093		1214	
	TIME	Times smaller than a day	evening	77944		120	
	QUANTITY	Measurements, as of weight or distance	100-foot	15736		142	
PLACES	GPE	Countries, cities, states	Korea	115406	22.86	6528	0.146
	LOC	Non-GPE locations, mountain ranges, bodies of water	the Middle East	43917		773	
	FAC	Buildings, airports, highways, bridges, etc.	San Ciprián	9393		384	
AGENTS	PERSON	People, including fictional	Tim Cook	771286	181.37	5741	1.125
	ORG	firms, agencies, institutions, etc.	Apple Inc.	555867		52714	
	NORP	Nationalities or religious or political groups	Chinese	11653		863	
OTHERS	PRODUCT	Objects, vehicles, foods, etc. (not services)	Watch Ultra	97619	16.07	3100	0.084
	EVENT	Named hurricanes, battles, wars, sports events, etc.	Super Bowl	10825		834	
	LAW	Named documents made into laws	the IRA Inflation Reduction Act	7372		76	
	WORK_OF_ART	Titles of books, songs, etc.	America's Greatest Workplaces 2024	2492		395	
	LANGUAGE	Any named language	English	315		7	

Table A.5: **Correlations of Sentiment Signals After Removing Different Categories of Entity Information**

This table shows correlations among sentiment scores extracted from raw transcripts (RAW), TRF-anonymized transcripts (TRF) and transcripts where one category of entities has been replaced (NUMBERS, PLACES, AGENTS or OTHERS). All sentiment measures are extracted using the GPT-4o-mini model. All values are Pearson correlation coefficients.

Entity Replacement Sentiment Correlation						
	$Sentiment_{RAW}$	$Sentiment_{TRF}$	$Sentiment_{NUM}$	$Sentiment_{PLC}$	$Sentiment_{AGT}$	$Sentiment_{OTH}$
$Sentiment_{RAW}$	1.000					
$Sentiment_{TRF}$	0.864	1.000				
$Sentiment_{NUM}$	0.876	0.899	1.000			
$Sentiment_{PLC}$	0.946	0.854	0.868	1.000		
$Sentiment_{AGT}$	0.912	0.863	0.872	0.917	1.000	
$Sentiment_{OTH}$	0.939	0.860	0.866	0.942	0.910	1.000

Table A.6: **Explanatory Power Comparison of Sentiment Scores Without Controls**

This table presents the results of univariate regressions with DGTW-adjusted returns on the earnings conference call day as dependent variables. The independent variables are sentiment measures extracted using GPT-4o-mini from earnings conference call transcripts. Columns (1) through (3) augment this baseline model by individually adding the sentiment scores derived from the four text sources. Columns (5) and Column (6) include both the sentiment extracted from raw transcript and from an anonymized transcript. All variables are winsorized at the 1st and 99th percentiles. All independent variables are standardized. All regressions include date fixed effects. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$				
VARIABLES	(1)	(2)	(3)	(4)	(5)
$Sentiment_{RAW}$	2.549***			1.842***	1.833***
	(19.602)			(8.741)	(7.237)
$Sentiment_{TRF}$		2.413***		0.820***	
		(17.329)		(3.766)	
$Sentiment_{LLM}$			2.410***		0.838***
			(17.521)		(3.212)
Date FE	Yes	Yes	Yes	Yes	Yes
Observations	7352	7352	7352	7352	7352
Adj. R^2	0.078	0.070	0.070	0.080	0.080

Table A.7: **Summary Statistics of Yahoo News Headlines**

This table presents the summary statistics for Yahoo news headlines from November 2023 to December 2024. Panel A provides an overview of the sample, including the total number of news headlines, firm-date level observations, as well as the number of firms and dates, along with the percentage of headlines posted before market close. Panel B presents detailed distributional statistics of both textual and firm-level characteristics. Specifically, it includes headline length (in tokens), the proportion of entities identified by the TRF model, and firm-level variables such as market capitalization (measured in millions of dollars), book-to-market ratio, and stock turnover.

Panel A: Sample Overview

Headlines	Firm-Date-Obs	Firms	Dates	Before Mkt Close (%)
52,716	30,950	2,052	292	64.66

Panel B: Textual and Fundamental Statistics

Statistics	Length	Entity%(TRF)	Size (Million $)	Book-to-Market	Turnover
Mean	15.55	38.30	121,493	0.50	2.99
Std. Dev.	6.13	20.45	419,148	0.50	9.63
25th Percentile	12.00	22.22	2,154	0.18	1.27
Median	14.00	35.71	9,275	0.38	1.85
75th Percentile	18.00	52.63	53,449	0.70	2.91

Table A.8: **Anonymization Gap and Information Loss**

This table examines whether the gap between sentiment scores from raw and anonymized transcripts is informative about anonymization-induced information loss in explaining contemporaneous stock returns. The dependent variable in all columns is the DGTW-adjusted stock return on the earnings-announcement date ($DGTW_{i,t}$). $Sentiment_{TRF}$ and $Sentiment_{LLM}$ denote the sentiment score computed from TRF- and LLM-anonymized transcripts. The variable Gap is defined as the absolute difference between the raw-text and anonymized sentiment scores, $Gap = |Sentiment_{RAW} - Sentiment_{Anonymized}|$. Each specification includes $Sentiment_{TRF}$, Gap, and their interaction $Sentiment_{TRF} \times Gap$. Columns (1)–(3) use sentiment scores generated by GPT-4o-mini, whereas columns (4)–(6) use scores generated by GPT-4o. The row "Period" indicates whether each regression is estimated on the pre- or post–knowledge-cutoff subsample. The row "Filter" reports whether the regression is run on the full sample ("–") or on a restricted anonymized subsample ("Anonymized"). In the anonymized subsample used in columns (3) and (6), we keep only transcripts for which, after TRF anonymization, the corresponding model (GPT-4o-mini in column (3) and GPT-4o in column (6)) fails to recognize the firm name, so as to better isolate anonymization-related information loss and mitigate forward-looking bias. Control variables are identical to those in Table 4. All variables are winsorized at the 1st and 99th percentiles, and all independent variables are standardized. All specifications include date fixed effects. T-statistics based on time-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $DGTW_{i,t}$					
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
$Sentiment_{TRF} \times Gap_{TRF}$	-0.388***	-0.576***	-0.460***			
	(-5.702)	(-7.585)	(-6.732)			
Gap_{TRF}	-3.284***	-4.029***	-3.205***			
	(-4.595)	(-5.214)	(-4.705)			
$Sentiment_{TRF}$	2.003***	2.517***	2.295***			
	(14.717)	(17.965)	(19.789)			
$Sentiment_{LLM} \times Gap_{LLM}$				-0.318***	-0.547***	-0.358***
				(-4.448)	(-5.750)	(-3.105)
Gap_{LLM}				-3.209***	-4.865***	-3.193**
				(-4.276)	(-7.271)	(-2.547)
$Sentiment_{LLM}$				1.945***	2.425***	1.915***
				(15.316)	(17.357)	(7.299)
Period	Pre	Post	-	Pre	Post	-
Sample	-	-	Identified	-	-	Identified
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,666	7,106	8,329	5,666	7,106	1,407
Adj. R^2	0.126	0.137	0.128	0.128	0.139	0.083

Table A.9: **Sample Periods and Predictive Accuracy of Downgrade Estimations**

This table presents the regression results predicting future S&P credit rating downgrades. The dependent variable is $Downgrade_{i,t+1}$, an indicator variable equaling 1 if S&P downgrades the firm's credit rating within the following 12 months from the earnings call release, and 0 otherwise. The key predictors, denoted as $\widehat{Downgrade}$, represent the predicted probability of a S&P credit rating downgrade within the 12 months following the earnings call release. Columns (1)–(3) report the results for Raw, TRF-anonymized, and LLM-anonymized transcripts respectively. The interaction terms with Pre denote the period before the knowledge cutoff, which equals 1 if the observations belong to pre cut-off date subsample, and 0 for post subsample. Control variables include $Leverage_{Asset}$, MTB, ROA, $RevGrowth$, $Std(Income)$, $Std(Revenue)$, $Zscore$ and $Oscore$. All specifications include SIC3 × Quarterly fixed effects. T-statistics based on SIC3-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $Downgrade_{i,t+1}$		
VARIABLES	(1)	(2)	(3)
$\widehat{Downgrade}_{RAW}$	0.028***		
	(5.808)		
$\widehat{Downgrade}_{RAW} \times Pre$	0.006		
	(0.644)		
$\widehat{Downgrade}_{TRF}$		0.023***	
		(5.350)	
$\widehat{Downgrade}_{TRF} \times Pre$		0.001	
		(0.116)	
$\widehat{Downgrade}_{LLM}$			0.025***
			(5.388)
$\widehat{Downgrade}_{LLM} \times Pre$			0.000
			(0.048)
Controls	Yes	Yes	Yes
SIC3 × Quarter FE	Yes	Yes	Yes
Observations	7,744	7,744	7,744
Adj. R^2	0.145	0.135	0.137

Table A.10: **Identification and Predictive Accuracy of Downgrade Estimations**

This table examines whether the identification of anonymized earnings call transcripts affects the predictive power of estimated downgrade probabilities for future S&P credit rating downgrades. The dependent variable is $Downgrade_{i,t+1}$, an indicator variable equaling 1 if S&P downgrades the firm's credit rating within the following 12 months from the earnings call release, and 0 otherwise. The key predictors, denoted as $\widehat{Downgrade}$, represent the predicted probability of a S&P credit rating downgrade within the 12 months following the earnings call release. $\widehat{Downgrade}_{TRF}$ and $\widehat{Downgrade}_{LLM}$ denote the predicted downgrade probabilities computed from TRF-anonymized and LLM-anonymized transcripts, respectively. $Identified_{TRF}$ ($Identified_{LLM}$) is an indicator variable equaling 1 if GPT-4o-mini can identify the date or firm name from the TRF-anonymized (LLM-anonymized) text, and 0 otherwise. Columns (1)–(2) use the TRF-based predictions, split into Pre and Post knowledge cutoff periods. Columns (3)–(4) repeat the analysis for LLM-based predictions. Control variables include $Leverage_{Asset}$, MTB, ROA, $RevGrowth$, $Std(Income)$, $Std(Revenue)$, $Zscore$ and $Oscore$. All specifications include SIC3 × Quarterly fixed effects. T-statistics based on SIC3-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Dependent Variable: $Downgrade_{i,t+1}$			
VARIABLES	(1)	(2)	(3)	(4)
$\widehat{Downgrade}_{TRF} \times Identified_{TRF}$	0.006	0.008		
	(0.695)	(0.854)		
$Identified_{TRF}$	0.006	-0.003		
	(0.885)	(-0.364)		
$\widehat{Downgrade}_{TRF}$	0.019**	0.017**		
	(2.082)	(2.324)		
$\widehat{Downgrade}_{LLM} \times Identified_{LLM}$			0.015	0.016
			(0.844)	(0.638)
$Identified_{LLM}$			0.002	0.019
			(0.157)	(0.926)
$\widehat{Downgrade}_{LLM}$			0.023***	0.023***
			(2.855)	(4.766)
Period	Pre	Post	Pre	Post
Controls	Yes	Yes	Yes	Yes
SIC3 # Quarter FE	Yes	Yes	Yes	Yes
Observations	3,341	4,352	3,341	4,352
Adj. R^2	0.133	0.142	0.135	0.144

B Prompts

B.1 Anonymization Prompt

We adopt the entity neutering methodology proposed by Engelberg et al. (2025), which employs two primary strategies: Masking and Paraphrasing. For each strategy, we utilize an initial prompt and an iterative prompt. The iterative prompt is triggered if the initial anonymization attempts fail to hide the entity's identity, employing more aggressive instructions.

B.1.1 Masking Prompts

The masking strategy replaces specific identifiers with generic placeholders. The initial masking prompt is defined as follows, where {NAME}, {TICKER}, and {INDUSTRY} are placeholders for the specific firm's details:

Your role is to ANONYMIZE all text that is provided by the user. The text you need to anonymize will be about the firm: {NAME} whose ticker is: {TICKER} and is in the industry of: {INDUSTRY}. Any instance of the firm, their products, the industry or industry related areas they are involved in should be anonymized. Absolutely no mention of key words related to that industry can be left in the text. After you have anonymized a text, NOBODY, not even an expert financial analyst should be able to use the text to identify the company or the industry the company operates in. For example, if the text is: The country's largest phone producer Apple had great phone-related earnings but Google did not in 2024, likely because of Apple's slogan Think Different, then you should ANONYMIZE it to: The country's largest product_type_1 producer Company_1 had great product_type_1 related earnings but Company_2 did not in time_1 likely because of Company_1's slogan slogan_1. You should also ANONYMIZE any other information which one could use to identify the company or to make an educated guess about its identity. Stock tickers are identifiers, are usually four or fewer capitalized letters (use TIKR as a stand-in for an arbitrary ticker), and can be referenced in the text in the following formats; SYMBOL:TIKR, $TIKR, >TIKR, $ TIKR, SYMBOL TIKR, SYMBOL: TIKR, > TIKR. Make sure you ANONYMIZE TIKR to ticker_x. Also ANONYMIZE any other identifiers related to companies, including the names of individuals, locations, industries, sectors, product names and types and generic product lines, services, times, years, dates and all numbers and percentages in the text, including units. These should be replaced with name_x, location_x, industry_x, sector_x, product_x, product_type_x, product_line_x, service_x, time_x, year_x, date_x and number a, b, c, respectively. Also replace any website or internet links with link_x. You should never just delete an identifier; instead, always replace it with an anonymized analog. After you read and ANONYMIZE the text, you should output the anonymized text and nothing else.

In the iterative masking phase, the prompt is modified to be more aggressive to ensure anonymity. The beginning of the prompt is replaced with: *"Your role is to ANONYMIZE all text that is provided by the user. You have already tried to anonymize the following text and failed, leaving details which were used to identify the text by an expert. Be more aggres-*

sive in your efforts to anonymize and pay more attention, so that you do not leave even the most minute details in the anonymized text."
Additionally, the instructions at the end are reinforced with: *"You have already failed me before, take no chances this time and be aggressive in censoring the text. After you read and ANONYMIZE the text, you should output the anonymized text and nothing else."*

B.1.2 Paraphrasing Prompts

The paraphrasing strategy rewrites the text to remove identifying writing styles and structures while preserving core information. The initial paraphrasing prompt is defined as follows:

You are to paraphrase text ensuring that its core information content is unchanged, such that someone who has memorized the text could not recognize it. Make sure to replace all unique or identifying words and phrases. Also change the sentence structure so that the resulting text is completely different from the original. This includes changing quotes and other highly recognizable structures in the text. The text provided will be about the firm: {NAME} whose ticker is: {TICKER} and is in the industry: {INDUSTRY}. Any information which could ever be used to identify the firm the text is about or when the text was written MUST be anonymized. Absolutely no mention of key words related to that industry or any other industries can be left in the text. If you observe any names, proper nouns, aliases, acronyms, dates, industry references (related and unrelated to the industry of the firm in question), government agency names, product names or types, etc., replace them with anonymous strings. Anonymity is the prime objective, leave zero hints for someone who has memorized the text! Ensure that the paraphrased text has approximately the same number of words as the original text. After you read and ANONYMIZE the text, you should output the anonymized text and nothing else.

Similar to the masking strategy, the iterative paraphrasing prompt is modified to strictly enforce anonymity. The beginning is modified to: *"You are to paraphrase text ensuring that its core information content is unchanged, such that someone who has memorized the text could not recognize it. You have already failed to paraphrase the text to make it anonymous. Be more aggressive in your paraphrasing and anonymizing of the text. Make sure to replace all unique or identifying words and phrases. Also change the sentence structure so that the resulting text is completely different from the original. This includes changing quotes and other highly recognizable structures in the text."*
The prompt concludes with the reinforced instruction: *"You have already failed once before, take no chances this time, be aggressive in your effort to anonymize the text. After you read and ANONYMIZE the text, you should output the anonymized text and nothing else."*

B.2 Anonymization Test Prompt

To verify the effectiveness of the anonymization, we use the following prompt to test whether an LLM can re-identify the firm, its industry, or the date from the anonymized text, following Engelberg et al. (2025):

*You will receive a body of text which has been anonymized. You are omniscient. Use all of your knowledge and any clues in the text to identify which company the text is about, its industry, and the date it was written. Provide your best guess if you are unsure. Your options for industry guesses are the following: Agriculture, Food Products, Candy & Soda, Beer & Liquor, Tobacco Products, Recreation, Entertainment, Printing and Publishing, Consumer Goods, Apparel, Healthcare, Medical Equipment, Pharmaceutical Products, Chemicals, Rubber and Plastic Products, Textiles, Construction Materials, Construction, Steel Works Etc, Fabricated Products, Machinery, Electrical Equipment, Automobiles and Trucks, Aircraft, Shipbuilding & Railroad Equipment, Defense, Precious Metals, Non-Metallic and Industrial Metal Mining, Coal, Petroleum and Natural Gas, Utilities, Communication, Personal Services, Business Services, Computers, Electronic Equipment, Measuring and Control Equipment, Business Supplies, Shipping Containers, Transportation, Wholesale, Retail, Restaurants Hotels & Motels, Banking, Insurance, Real Estate, Trading, Almost Nothing. Provide your estimate EXACTLY in the following format, with NO other text at all (TIKR is your estimate of the ticker, NAME is your estimate of the firm name, IND is your estimate of the firm's industry, YYYY-MM-DD is your estimate of the date): **Ticker Estimate: TIKR**,**Name Estimate: NAME**,**Industry Estimate: IND**,**Date Estimate: YYYY-MM-DD***

B.3 Sentiment Prompt(For Transcript)

In designing the prompt for our analysis, we reference the methodology for controlling LLM output tokenization from Engelberg et al. (2025). They demonstrate that by forcing the model to generate a number in a structured format—such as 0.X for a sentiment score—one can isolate the magnitude of the estimate into the log probability of a single, unambiguous token (the digit X). We adopt this principle to ensure that the numerical values generated by our prompt are free from tokenization artifacts, such as leading or trailing spaces. This structured approach allows us to reliably use the model's log probabilities as a direct measure of its assessment. We use the following prompt:

*You are an expert Financial Analyst. Your task is to read quarterly company earnings call transcripts and infer if the information in the transcripts is a good or bad signal for the future returns of the security of the company whose earnings call is being transcribed. Based on your answer, I will choose to buy or sell that security. After reading the transcripts, provide a bull or bear signal as two numbers: the direction and the magnitude. For direction, output only 0,1, or NA, where 0 is bearish, 1 is bullish, and NA means there is no information relevant to security prices. For magnitude, output a number ranging from 0 to 1 ONLY, where numbers near 0 indicate slightly bearish (if the direction is 0) or slightly bullish (if the direction is 1), and numbers near 1 indicate highly bearish (if the direction is 0) or highly bullish (if the direction is 1). Provide your output exactly in the following format, with no other text at all: **Direction Estimate: DIRECTION**,**Magnitude Estimate: MAGNITUDE**.*

B.4 Uncertainty Score Prompt

We refer to the definition in Loughran and Mcdonald (2011) to extract the uncertainty score from the transcript, with the following prompt:

*Analyze the following conference call transcript. As a financial analyst specializing in textual analysis, your task is to evaluate the level of Uncertainty. Your evaluation must be on a continuous scale from 0 (representing absolute certainty) to 1 (representing maximum uncertainty) ONLY. This assessment should not be based on a simple count of keywords. Instead, you must analyze the contextual meaning, usage, and significance of words and phrases that convey a cautious or uncertain tone, emphasizing the general notion of imprecision rather than exclusively focusing on risk. Your goal is to determine the genuine, unmitigated uncertainty conveyed by the transcript. Provide your output only in the following format, with no other text: **Uncertainty Score: UNCERTAINTY**.*

B.5 Investment Score

We refer to the prompt in Jha et al. (2024a) to extract the Investment score from the transcript, with the following prompt:

*The following text is a company's earnings call transcript. You are a finance expert. Based on this text only, please answer the following question. How does the firm plan to change its capital spending over the next year? There are five choices: Increase substantially, increase, no change, decrease, and decrease substantially. Please select one of the above five choices for each question and provide a one-sentence explanation of your choice for each question. The format for the answer to each question should be **choice - explanation**. If no relevant information is provided related to the question, answer **no information is provided**.*

B.6 Economy Score

We refer to the prompt in Jha et al. (2024b) to extract the Economy score from the transcript, with the following prompt:

*The following text is a company's earnings call transcript. You are a finance expert. Based on this text only, please answer the following question. Over the next quarter, how does the firm anticipate a change in optimism about the US economy? There are five choices: Increase substantially, increase, no change, decrease, and decrease substantially. Please select one of the above five choices for each question and provide a one-sentence explanation of your choice for each question. The format for the answer to each question should be **choice - explanation**. If no relevant information is provided related to the question, answer **no information is provided**.*

B.7 Sentiment Prompt(For News)

We used a similar LLM prompt to extract sentiment from news:

*You are an expert Financial Analyst. Your task is to read news headlines and infer if the reported headline is a good or bad signal for the future returns of the security being written about. Based on your answer, I will choose to buy or sell that security. After reading the headline, provide a bull or bear signal as two numbers: the direction and the magnitude. For direction, output only 0,1, or NA, where 0 is bearish, 1 is bullish, and NA means there is no information relevant to security prices. For magnitude, output a number ranging from 0 to 1 ONLY, where numbers near 0 indicate slightly bearish (if the direction is 0) or slightly bullish (if the direction is 1), and numbers near 1 indicate highly bearish (if the direction is 0) or highly bullish (if the direction is 1). Provide your output exactly in the following format, with no other text at all: **Direction Estimate: DIRECTION**,**Magnitude Estimate: MAGNITUDE**.*

B.8 Downgrade Prediction Prompt

We adapt the approach of Donovan et al. (2021) to design a prompt that extracts the predicted probability of an S&P credit rating downgrade within one year of the earnings call:

*You are an expert Credit Risk Analyst. Your task is to to read quarterly company earnings call transcripts and estimate the probability that the company will experience an S&P credit rating downgrade within the next twelve months from the date of this call. Based on your analysis, output only a single probability score between 0 and 1, where a score near 1 indicates a very high likelihood of a downgrade (severe credit risk) and a score near 0 indicates a very low likelihood (stable or improving credit profile). Provide your output exactly in the following format, with no other text at all: **Downgrade Estimate: [score]**.*